

**EPS 210:**  
**DEVELOPMENTAL PSYCHOLOGY OF EDUCATION**  
**LECTURER: DR. SUSAN N. NGUNU**  
**DEPT EDUCATIONAL PSYCHOLOGY**

# Definition of Developmental Psychology, Growth and Development

- **Meaning – Developmental Psychology**
- The study of how humans grow, develop, and change throughout the lifespan.
- It studies the processes and changes involved in physical, mental(cognitive), social, emotional and psychological changes that occur in the individual from conception to adulthood.
- **Human Growth and Development:** These two terms have been used interchangeably by many scholars including educational and developmental psychologists because the two processes are interrelated and interdependent.
- They can be **distinguished as follows**

# Meaning – Human Growth

- It generally refers to increase in some quantity over time. E.g. size, length, height and weight
- Such changes could be objectively observed and measured

# Characteristics of Human Growth

- Refers to increase caused by becoming larger and heavier.
- It is quantitative, additive, and incremental.
- It is objectively observable and measurable.
- Occurs rapidly in early phases of life up to adolescence.
- It may or may not bring development.
- Rate of growth is not uniform. Brings about individual differences among children.
- Different body parts grow at different rates.



# Meaning – Human Development

- Refers to changes over time in a person's body, thought, and behavior due to biological and environmental influences.
- Results in **improved working or functioning**.
- Indicates the changes in the quality or characteristics.
- Results from growth, maturation and learning. E.g. How does a child's movement develop?

# Characteristics of Development

- Occurs in an orderly, predictable pattern as a result of maturation and experience/learning.
- Is continuous throughout the lifespan.
- Aims at adapting to the environment.
- Is complex involves many structures and functions.
- Has many aspects (physical, emotional, intellectual, social, and moral).
- Is qualitative thus cannot be measured directly.
- Is possible even without growth.

# Distinction between growth and development

- **Growth**

- 1 Quantitative
- 2 Objectively observed & measured
- 3 Fixed. Stops after maturation.
- 4 It is one aspect of development
- 5 Occurs in different body parts.
  - i.e. it's organic
- 6 It is not directional.
- 7 It is not affected by learning.
- 8 May or not lead to development.
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- **Development**

Qualitative

- It cannot be measured directly.
- Continues throughout lifespan.
- It is complex and many sided.
- It involves changes in processes and functions.( non-organic)
- It is progressive and sequential.
- Affected by Learning and experience.
- It is integrative. Dev't in one aspect promotes the dev't in other aspects.

# Domains of human development

- Human development is studied within four interacting domains:
  - a. physical growth,
  - b. cognitive growth,
  - c. personality development, and
  - d. sociocultural development

# Aspects of human development

- Physical: weight, body proportions and general physical appearance
- Affected by biological influences like genes from parents, and maturation
- Cognitive/Intellectual development- Acquisition of skills in perceiving, thinking, reasoning, problem solving, and language reflects the role of cognitive processes in development.

# Aspects of human development continued

- Emotional development- involves changes in an individual's emotions
- Personality development- Acquiring stable and enduring personality traits
- Social development- These are changes in an individual's relationship with others through the processes of socialization and enculturation

# Principles of growth and development

- Several principles of development have been put forth from the scientific knowledge on children.
- **Development is orderly**
- States that development proceeds from general to specific and from simple to complex
- E.g. Young children stand before walking, and utter single words before forming complex sentences.

# Principles of growth and development continued

- **Development is sequential**
- Development follows an orderly, sequential, irreversible, and predictable pattern. It is not random.
- The sequence of physical and behavioural stages is quite consistent.
- E.g. Puberty is achieved before full physical structure.
- A child sits, crawls, stands, and then walks.
- Disease or severe malnutrition may halt its continuity for some time



# Principles of growth and development

## Continued

- **Development is a cumulative and continuous process**
- This principle emphasizes that developmental changes are cumulative and they occur in a bit-by-bit fashion.
- Such changes proceed from birth to death.
- Developmental change is not always smooth and gradual. There are periods of accelerated growth. i.e. growth spurts.

# Principles of growth and development continued

- **Critical and Sensitive periods**
- There are critical and sensitive periods
- **Critical periods**- the time periods of most sensitivity to environmental influences.
- These are periods when the potential for growth as well as harm from the environment is and its maximum.
- Any interference could result to permanent deficiencies or malfunction

# Principles of growth and development continued

Development is directional and uniform.

It follows two trends:

- a. **Cephalo-caudal trend**– development proceeds from head to tail. i.e, at any specific time, structures nearer the head are more developed than those near the toes.
  - E.g. A child can hold the head upright and use hands before walking or using the legs.
- b. **Proximodistal trend**- development proceeds from centre-line of the body to other parts. E.g. The heart develops faster than the hands, child learns to wave before learning to pick using fingers.

# Development is a product of both maturation & learning

- Maturation- the changes that occur according to one's genetically determined, biological timetable of development.
- Learning- A relatively permanent change in behaviour, capability, or attitude due to experience/training.
- The two interact to influence growth and development.
- E.g. One develops the capacity to read as one's cognitive ability increases and due to exposure to reading experiences and training.

# Development results from the interaction of heredity & environment

- An individual is the product of both heredity and environment.
- Genes- determine the rate and quality of growth and development (they lay the genetic potential/blueprint).
- Environment- (Socio-cultural, emotional and physical) determine how the potential unfolds/is realized.

# Development is unique and individualized process.

- Even with a set pattern of development, every individual differs regarding the achievement of the changes.
- Individual differences are the unique characteristics that distinguish a person from other members of a larger group.
- These differences could due to genetic factors, and environmental ones like: nutrition, illness, lack of exercise, etc.
- E.g. All children crawl before standing but the age at which they perform these activities is different.

# Principles of growth and development continued

- **Development involves differentiation**
- All behaviour becomes more highly differentiated and adapted with development.
- E.g. The many organs of the body (e.g., the brain, heart, and lungs) have their origin in a single cell.

# Principles of growth and development continued

- **Development is inter-related**
- Different aspects of development are inter-related i.e different aspects of development take place in an individual and influence each other: social, mental, and emotional.
- These three are inter-related and one doesn't exist without the other.
- Development is influenced by one's cultural context, historical time and lifetime experiences.



# Principles of growth and development continued

- **Development is holistic**
- Development involves large and significant individual differences in all qualitative aspects of behavioural development. It does not occur in piecemeal.

# Principles of growth and development continued

- **The principle of plasticity**
- This principle states that individuals have the capacity to change in response to positive or negative life experiences.
- Plasticity allows recovery from early deprivation and for optimal adaptation to diverse environmental circumstances.

# Importance of the studying of human growth and development/Developmental Psychology

- Knowledge of development patterns, what they are like and what causes variations in the development of children, is essential for both scientific and practical reasons. It will;
- The study provides a rich background of information about children's behaviour and psychological growth under a variety of conditions. The teacher being an agent of socialization is expected to cultivate desirable changes in the behaviour of children using the information.
- Help the teacher to understand the developmental stages of the child and the characteristics that emerge at different age levels so that he can correctly cater for the needs of the learners at these stages.

# Importance of the studying of human growth and development

- Helps the teacher understand the causes of individual differences among learners and how to cater for them.
- Helps the teacher understand the factors that are responsible for the mental ill health and maladjustment in children.
- Plays a predictive role ie helps one to approximate at what age certain behavioural changes take place and when these patterns are generally replaced by more mature patterns. This is significant since, if too much is expected from children, they develop a feeling of inadequacy. On the other hand if too little is expected of them, they do not have an incentive to realise their potential.

# Importance of the Developmental Psychology

- Knowing exactly what to expect from children enables developmental psychologists, teachers and parents to set guidelines in the form of height-weight scales, mental age scales etc. Deviations from normal development may be investigated and appropriate intervention given. It be planned to help those who vary considerably in personal, social and emotional development.
- Helps teachers and parents guide the child's learning properly. A child must be helped to acquire skills of walking when he is of an age appropriate for this skill. Not providing learning opportunities at the appropriate time may delay the normal development.
- When the development pattern is normal, children are effectively prepared for the next stage in life.

# Importance of the studying of developmental psychology

- Knowledge of development patterns helps teachers and parents prepare the child psychologically for the physical and behavioral changes that will occur as they grow up/ prepare them to be good citizens
- Helps the teacher to understand past aspects of development which may have a bearing on the students present life ( a student is usually a product of what they have gone through)
- Helps teacher to understand himself/herself better

# HISTORICAL PERSPECTIVE OF THE STUDY OF DEVELOPMENTAL PSYCHOLOGY

- Before the development of child psychology as a scientific discipline, people's understanding of the child was influenced by the prevailing philosophies of their time. such as;
- **PLATO 428-348BC**
- He advocated that child rearing should be done by the **state**.
- A child **should not be spoilt** but should also **not be subjected to tyranny**.
- He argued that children should be reared to become **constructive members of society**.
- Both boys and girls should be provided **equal opportunities** and should learn to govern themselves.

# Historical Perspective of the Study of Developmental Psychology Cont'd

- **ARISTOTLE 384-322 BC**
- Unlike Plato, Aristotle argued that children should have opportunities for **individual liberty and privacy**.
- The **family should raise children** so as to provide **personal and social stability**.
- He expressed concern about individual differences.
- For the family to function properly, females must devote themselves to rearing and supervision of the home.



# THE DARK AGES 4-14<sup>TH</sup> C AD

- Also called the **period of inherent wickedness**
- This was after the fall of the Athenian Civilization. Power and culture shifted to Rome hence the term “dark ages”.
- The leaders (monks) became responsible for matters pertaining to behaviour and development.
- The church held that human beings were born sinful and wicked.
- Parents were told that misbehaviour of their children was due to innate wickedness i.e. original sin.
- A child engages in negative behaviour because his inherent sinfulness had not been sufficiently suppressed.

# THE DARK AGES 4-14<sup>TH</sup> C AD

- Therefore the goal of child-rearing at that time was salvation and the puritans in particular considered it necessary to stamp out sin in children.
- Children had to be subjected to hard work and constant severe punishment.
- There were extremes i.e. happiness was seen as a sign of the devil, playfulness and cheerfulness were signs of damnation and had to be stamped out right from the birth.

# FROM WICKEDNESS TO INNOCENCE (REINAISSANCE) 1200-1700 AD

- From 1200 and 1600 AD , the **child was of very little significance.**
- Many children died at infancy.
- **Ignorance** was widespread among the rich and the poor.
- **Breastfeeding** was considered **vulgar.**
- Nothing was known about germs or inoculations because medical treatment had little scientific basis.

# 1200-1700 AD FROM WICKEDNESS TO INNOCENCE (REINAISSANCE)

- Due to the **high rate of mortality**, parents developed an **unsympathetic and fatalistic attitude** towards children.
- They did not view the child as an individual until they were sure he or she would survive.
- This changed by the beginning of 17<sup>th</sup>c. Childhood started being recognized as a unique and crucial period of human life.

# 1200-1700 AD FROM WICKEDNESS TO INNOCENCE (REINAISSANCE)

- Parents and teachers were advised to devote much more care and thought to child rearing and education.
- The innocence of childhood was to be safeguarded and strengthened.
- Those that shaped the thinking during this period were;

# JOHN LOCKE (1632-1704)

- English philosopher and physician (doctor).
- He reckoned that early childhood experiences could have profound effect on later adulthood.
- Opposed the idea that children are innately sinful.
- Proposed that newborns were like a blank slate (tabula rasa) upon which experiences from the outside world as well as adults were written on.
- Childhood was responsible for later behaviour and formation of lifelong habits and therefore deserved a great deal of attention.

# JOHN LOCKE (1632-1704)

- He argued that parents should:
- Teach children socially desirable habits, to reason, help them control their appetites and impulses as this would help them to grow into rational and responsible members of society.
- Encourage expression of the children's natural curiosity because children can only become rational if they learned to express their curiosity.
- Provide encouragement by answering their children's questions with attention and respect.
- Behave in a rational way so as to command children's respect, attention, and affection.
- Use praise and commendation instead of punishment.

# JEAN JACQUES ROUSSEAU

- He was a French philosopher.
- Believed that the child is born with **an innate moral sense** i.e. he has an assumption that children are inherently **good and noble**.
- Children should be observed carefully and be given the kind of education which fits individual personality and characteristics at different stages of development.
- Parents should not use harsh methods to discipline a child.
- They should instead create a desire in the child to learn.



# BABY BIOGRAPHERS

- In the late 18<sup>th</sup> C, a few curious and courageous individuals attempted to learn about children through actual observation.
- They were able to provide a detailed record of an infant's growth and development over a period of time. These included;
- **Tiedmann** (1787)- a German published a book tracing the sensory, motor, language and intellectual growth of a single infant during the first 2 ½ years of life.
- **Johann Heinrich Pestalozzi** (1746-1827)- from Switzerland. He published notes based on the careful observation of the development of his 3 ½ year-old son

# BABY BIOGRAPHERS

- **Charles Darwin (1877)**-kept a diary called “baby journal” about his infant son that he began in 1840 but did not publish until 30 years later
- **Millicent Shinn (1900)**-from USA. She published notes based on her observation of her niece from birth through the first year of life.

# BABY BIOGRAPHERS

- **Limitations**
- Baby biographies **did not** provide good scientific data for the following reasons;
- The observations were unsystematic and made at irregular intervals
- The observers were relatives and selective in their perceptions i.e. they mostly saw and reported the positive, praiseworthy aspects of early development and neglected the negative aspects (possibility of halo effect i.e. not objective accounts)
- Since each account is based on only one case/child, it is impossible to make generalizations about all children from them.

# CHILD STUDY MOVEMENT

- This was the beginning of Child Psychology as a scientific study.
- The **baby biographers** aroused a lot of growing scientific interest in children in the 19thC. They were therefore the forerunners of the developmental and experimental traditions in child psychology.
- **Stanley Hall** founded the first child study movement that intended to explore the evolution of children's ideas through the process of recapitulation i.e. the child in his/her development repeats the history of the human race.
- Although the movement died due to attacks by Scientific methodologists, it nevertheless brought to the public attention, the need for scientific data on children

# THE CHILD STUDY INSTITUTES

- The **child study** movement paved the way for the establishment of child study institutes at major universities in the USA.
- Mrs Cora Bressey Hills established a **child research institute in 1906 and 1907**
- 1916-the IOWA child welfare research station was established. **It had 3 functions;**
  - Research
  - Teaching
  - Dissemination of information on children
- After WWI, the public became more aware of the problems of children especially the orphans i.e. illiteracy and poor health.

# THE CHILD STUDY INSTITUTES

- Lawrence Frank, an Economist got support from many foundations amongst them Rockefeller Memorial Fund (called Foundation today) and founded a child development Institute at Teachers' college of Columbia in 1924
- Institute of child welfare at University of Minesota in 1925
- The institute for child development at Yale in 1928
- By 1930, research in child development was in top gear and there were graduate programmes in it( masters, PhDs)

- The graduates of such programmes taught in most of the college and universities in USA
- Child development finally emerged as a psychological discipline
- There were varied beliefs about how children grew and what shaped their behavior/personality.

# 19<sup>TH</sup>C ONWARDS

- **J.B. WATSON (1920) -SHAPING HUMAN BEHAVIOUR**
- He was a behaviourist and advocated that psychology should deal with overt behavior.
- He applied Pavlovian principles of classical conditioning to human infants
- He asserted that by arranging sequences of conditional responses, he could control behavior in almost limitless ways
- He believed that he could train a child to be any kind of individual (Give me a dozen children and conditions of my own and I will create anything out of them so genes not vital)



- **BF SKINNER-SHAPING BEHAVIOUR MORE EFFECTIVELY**
- He was concerned about operant conditioning. He said that behavior that is truly reinforced will occur again and one that is not reinforced will diminish.
- He argued that personality development is a product of environmental experiences and that those who control reinforcement shape behavior.
- Behavior should therefore not be left to chance.

- **ABRAHAM MASLOW**

- The main theme of his work was “**self-determination**” captured in the slogan ‘trust children, let them grow’
- He believed that a person possessed an essential inner nature (inherited natural qualities) and ones’ behavior was therefore not dominated by external forces.
- He said that children like adults are able to make wise choices when their need for physical comfort, safety, love, care, belonging, acceptance by others and self-esteem needs are satisfied by others

# Recent Developments Related to the needs of the Child (20<sup>th</sup>c)

- **1924-Geneva Convention** on the child was held which passed that mankind owes to the child the best that it can give.
- **1946- United Nations (UN)** established UNICEF-United Nations International Children's Emergency Fund to cater for children who suffered after WWII. It was initially a 3 year plan meant for Europe but was expanded to the developing countries in 1950.
- 1953- it became an indefinite organization. The words International and emergency were officially dropped and became **United Nations Children's' Fund** although all the original initials UNICEF were retained.

# Recent Developments Related to the needs of the Child (20<sup>th</sup>c)

- 20<sup>th</sup> Nov 1959, the UN declared the rights of the child.
- 1979 was declared the international year of the child
- 1989 a convention of the rights of the child was held and ratified in 1990
- 29-30 Sept 1990 the largest gathering of world leaders in history assembled at the UN HQs to attend a world summit of children. 71 heads of state and governments and 88 senior officials (ministers) attended.
- The summit adopted a declaration on the survival, protection and development of children and a plan of children for implementing declaration in the

# Un Declaration of the Rights of the Child

Every child has the right to;

- Affection, love and understanding
- Full opportunity for play and recreation
- A name, family and nationality
- Free education
- Be among the first to receive relief in times of disaster

# Un Declaration of the Rights of the Child

- An opportunity to learn to be useful members of society and to develop individual abilities.
- Be brought up in a spirit of peace and universal brotherhood.
- Adequate nutrition, medical care and clothing
- Special care if handicapped
- Enjoy all these rights regardless of race, colour, sex, religion, nationality or social origin.

- **In Kenya**, several NGOs are concerned with alleviating children's' problems e.g.;
- UNICEF (researches on survival and development)
- Child Welfare Society
- Undugu Society of Kenya that rehabilitates street children
- Save the Child
- ANPPCAN-African Network for the Prevention and Protection against Child Abuse and Neglect
- Nacece/Dicece- district national centres for early childhood education.
- Kenya Alliance for advocacy of children's' rights.

- In March, 1990 a world summit was held in Thailand and the main discussion centred on provision of basic education for all children.
- June 1992-The Earth Summit held in Rio de Janeiro where agenda 21 chapter 25 urged governments to;
- Implement programmes to reach goals set by the world summit for children.
- Ratify and implement the convention on the rights of the children.



- Promote primary environmental care activities to improve the environment by meeting basic needs and empowering local communities.
- Expand children's education especially the girl child
- Incorporate children's concern into all relevant policies and strategies of environment and development.

# HUMAN NEEDS/ DEVELOPMENTAL NEEDS

- A **need** is an internal feeling of dissatisfaction/deprivation a person feels in his body when deprived of something. Deprivation will arouse an organism to replenish its stores. There are two kinds of needs;
- Physiological
- Psycho-sociological

- **Physiological needs**

- These are aimed at satisfying the basic survival needs of the organism.
- They are also called primary, biological, deficient needs;
- **Need for food**-necessary for body building, protection against diseases as well as repair of injuries.
- Serious malnutrition affects brain development, physical and intellectual functioning.
- Children who are poorly fed are more exposed to diseases, infections and emotional frustration.
- They also lack concentration, are usually tired, sleepy, easily irritable, restless, bored etc.
- Teachers and parents should ensure children get balanced diets as well as adequate amounts that are well spaced for proper growth and development.

- **Need for air**-adequate, fresh air/oxygen is needed for proper development.
- Poorly ventilated and overcrowded rooms lead to accumulation of foul air which makes children sick, tired.
- Dull, faint, irritable and tense.
- Homes and schools should provide spacious well ventilated rooms for children.

- **Need for clean water**-lack of water for long periods depletes body tissues of fluids which are required to maintain the proper water-sodium balance (Morris, C.G. 1982).
- Prolonged thirst especially after games could make children tense, irritable, fidgety.
- Dirty water can transmit water borne diseases such as typhoid, cholera.
- Teachers and parents should ensure that children get access to clean water especially in dry areas.

- **Need for sleep, rest and activity**-adequate sleep and rest is necessary for proper development of young children (6hrs for men, 7 hrs for women, 8 hrs for children, more than 8 hrs for fools).
- Lack of this could lead to being non-co-operative, inattentive, irritable, hatred for homework etc.
- Parents need to be advised on the necessity of ensuring that children get enough sleep.
- Teachers should be sensible in giving homework.
- They should also make lessons interesting to avoid boring the children.

- **Activity/play** is necessary for the following reasons;
- Physical development- It keeps the children physically fit through development of muscles.
- Social development-children learn many social skills e.g. give and take during play
- Language development-imitates sounds from others
- Intellectual development is stimulated through exploration, creativity, problem solving, increases attention span etc

- **Need for elimination** of waste products from the body-a child who has irregular elimination and suffers constipation is usually irritable, physically ill and may not concentrate in school. Schools should provide proper sanitation facilities.
- **Need for sex**-not for children! This is a primary need only if used for procreation to ensure survival or continuation of species. It is a strong urge that comes due to hormonal secretions. Social learning determines its expression. To prepare children for responsible sexuality and prevent sexual maladjustments such as promiscuity and over-indulgence later in life, sex education programmes that include biological, moral, emotional as well as social effects of sex abuse should be provided.



- The need for self-control and self-discipline as well as sublimation methods should be encouraged to delay gratification until the appropriate time eg art, drama, sports, proper use of leisure time.
- Sublimation is diverting/ channeling sexually aggressive energy into socially acceptable/ admirable energy(Mike Tyson and his boxing career)

- Other needs are **warmth and proper temperature, avoidance of pain and injury.**
- NB: if one of the physiological needs is not met, it will completely dominate the person (and prevent him from undertaking other tasks) till it is satisfied

# Psycho-sociological/Psycho-social needs

- Also referred to as sociogenic needs, learned, secondary, personality needs and include;
- **Need for affection**-need for love and a sense of belonging. This need is vital for the healthy development of a child's personality.
- It is met by the child experiencing a stable, continuous, dependable and loving relationships with parents, siblings and others around him.
- Through love/affection, the child realizes his personal identity and worthiness.
- This need is the basis of all later relationships with family, friends, colleagues and eventually the family that one raises.

- **Need for security**-Security to a child comes as a result of environmental and family stability.
- The attitude and behavior of adults should be socially acceptable, consistent and dependable to ensure continuity and predictability of the world as the child perceives it.
- Homes with definite standards of behavior set by parents help children to develop a sense of security.
- Consistency of school values helps the pupils to know and understand what sort of behavior is expected from them.

# Need for new experiences and intellectual stimulation

- New experiences are a prerequisite for mental growth.
- These range from grasping, crawling, walking, running, climbing etc.
- New experiences enable the child to learn basic lessons early, develop mastery.
- This gives the child joy and a sense of achievement and competence .
- It enhances, exploration, attention span, problem solving etc.
- Too much stimulation can lead to withdrawal and fear while too little can give rise to boredom and apathy hence balance needed.

# Need for praise and recognition-

- This helps the child to grow to be a self-reliant adult (from his initial helplessness) by developing his emotional, social, moral and intellectual capabilities.
- Parents should therefore appreciate what the child has achieved through praise, recognition and encouragement.
- This will make the child feel he is loved and accepted and will in turn make him try to please those that love him.
- Mistakes should be regarded as necessary and sometimes an integral part of learning.
- Recognition of the basic worth of the child fosters confidence within the child.

# Need for responsibility

- This is met by allowing the child to gain personal independence.
- This begins by allowing the child to look after himself in matters of everyday care such as feeding, dressing, washing himself etc. as the child grows older, independence should be increased.
- The child should have freedom of physical movement, of food, play, clothes, friends, studies, hobbies, career and eventually a marriage partner.
- Towards this end, parents and teachers should guide the child in the right path of freedom and explain the social values that must be transmitted.
- If values are imposed without explanations, the child may not take them seriously.

# REVISION QUESTIONS

- Describe the contributions of the following to the growth of Developmental Psychology:
- Baby biographers
- Child study institutes
- Early philosophers (Plato and Aristotle)
- What changes in the 20<sup>th</sup> Century have shaped the study of Developmental Psychology?
- How can parents, teachers, and the community provide for the various children rights in Kenya?
- What challenges are encountered in the provision of children rights in Africa?



# **PRE-NATAL DEVELOPMENT**

- **This is the development that occurs between conception and birth.**
- **Prenatal development consists of three stages.**
- **Each stage is marked by a specific event at the beginning and another at its end.**
- **The three stages are:**
  - **a. Germinal/zygote stage**
  - **b. Embryonic stage**
  - **c. Fetal stage.**

# Germinal stage (0-2 weeks)

- The period starts at conception and ends with the process of implantation, when the developing zygote becomes firmly attached to the wall of the uterus.
- **Fertilization**
- The egg is released into fallopian tube (ovulation).
- Sperm travels up the fallopian tube and meets the egg.
- Individual starts to form when the male and female reproductive cells (gametes) combine to form a single cell called a zygote.
- This process is referred to as *conception/fertilization*.

- This is facilitated by **meiosis**-the process of gamete formation which involves halving the number of chromosomes from 46 to 23.
- A zygote has 23 pairs of chromosomes.
- 22 are called autosomes ( somatic cells) and contain most of the genetic information for the new individual.
- The 23rd pair, the sex chromosome ( germ cell-ovum/ sperm ) determines the sex.

# Germinal Period cont...

- One begins life as a single celled organism, and develops into a human being made up of trillions of cells
- *Mitosis takes place after the zygote has been formed.*
- By mitosis it divides into two equal cells each with 23 pairs of chromosomes as in the original

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- These two cells and all resulting cells continue to divide, forming a ball-like structure called the blastocyst. Cell differentiation begins and the inner layer of the blastocyst becomes the embryo.
  - The outer layer of cells develops into tissues that protect and nourish the embryo.
  - The zygote moves from the fallopian tube toward the uterus for implantation.

# EMBRYONIC PERIOD- WEEK 3 - WEEK 8



Photo by Lunar Caustic

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- It starts when the zygote attaches itself to the uterine wall and ends when the first bone cells form
  - *Once the zygote attaches to the wall of the uterus, it is known as the **embryo**.*
  - During the **embryonic stage** *the major internal and external organs are formed.*
  - The inner layer of embryonic cells is busy forming the embryo itself,

## EMBRYONIC PERIOD- WEEK 3 - WEEK 8

- The blastocyst outer layer rapidly forms into four major support structures that protect and nourish the developing organism.
- i) **The amnion-** It is a water- tight sac that fills with amniotic fluid from the pregnant woman's tissues. It protects the embryo from damaging movements e.g. falls or blows, regulates its temperatures and provides a weightless environment that will make it easier for the embryo to move.
- ii) **Chorion:** It surrounds the amnion and eventually becomes the **lining** of the placenta.





- **iii) placenta**

- The placenta is the unborn baby's life support system and plays a key role in its development.
- It connects the mother to the fetus through the umbilical cord and carries out the following functions.
- Provides the baby with oxygen and nutrients.
- Passes oxygen from the maternal blood to the foetal blood, and carbon dioxide passes from foetal blood to maternal blood.

# Embryonic

- Removes harmful waste from the baby. They are filtered out by the kidneys of the mother.
- Produces hormones that help the baby grow.
- Passes immunity from the mother to the baby.
- Placenta functions as an efficient barrier and protect the fetus from infections (bacterial) and maternal diseases.
- The placenta stores glycogen, fat etc. for the foetus before liver is formed

## **Umbilical cord:**

- The umbilical cord *links the embryo directly to the placenta and transfers all material to the fetus.*
- It transports carbon –dioxide and metabolic waste from the embryo.

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- While the **outer layer** is forming the placenta and other membranes, **the inner cell mass** evolves into organs
- The cell mass separates into three layers:
- a. **Ectoderm** – outer layer which forms the basis for the development of skin, nails, teeth, eye lens, inner ear, and nervous system
- b. **Endoderm** – inner layer which forms the basis for the formation of digestive system, liver, pancreas, salivary glands, respiratory system, and lungs
- c. **mesoderm** – middle layer which forms the basis for the formation of skeleton, dermis or inner layer of skin, muscles, circulatory system and excretory system

- 
- **Organogenesis takes place –This is the process of organ formation during first two prenatal months.**
  - By week 3, the neural tube, which eventually becomes spinal cord, starts to form.

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- 
- By the 4<sup>th</sup> week, the heart begins to beat and circulate blood (enhances waste removal and acquisition of nutrients).
  - Legs become more differentiated, face starts to form, and intestinal tract appears.
  - By the 7-8wks, male embryos develop testes and female embryos develop ovaries.
  - The embryo already resembles a human being with limbs, fingers, toes, and many internal organs begin to function.



# Embryonic period cont....

- **N. B** Embryonic stage is the most critical stage of development since all major organs develop at this time and these organs are most vulnerable to harmful effects of substances in the prenatal environment (teratogens)
- Embryos that are severely deformed do not survive.
- A spontaneous abortion commonly called a miscarriage expels from the uterus an embryo or foetus that is defective and therefore unable to survive outside the womb.
- Most miscarriages occur at this stage due to abnormal pregnancies while 50% -70% involve chromosomal abnormalities.

# Fetal Period (Week 9-40)

- The final and longest period of prenatal development.
- It begins when bone cells begin to form, until birth.
- The developing organism is now called **foetus**.
- The foetus experiences rapid growth and continued development and differentiation of structures, organs and systems that emerged during the embryonic period.
- All the major organs and systems begin to function.



- 
- The fetus begin to move, sense and behave.
  - Coordination between the nervous and muscular system allows the fetus to kick its legs, make fists, twist its body etc

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- At the 12week -Sexual differentiation is progressing rapidly and sex of the fetus can be detected
  - At 16wks the fetus begins to suck, swallow, munch, hiccup, breath, cough thus preparing for extra uterine life.
  - At 17 to 18 weeks Emergence of eyebrows, eyelashes and scalp hair.
  - Skin thickens and is covered with vernix, a thick greasy substance, that protects the foetus in its stay in the amniotic fluid.

- 
- **7 months.: Is referred to as the age of viability-** age at which foetus can survive outside the uterus because most of its body systems function adequately.
  - Brain and eye development proceed very fast.
  - Foetus develops bone and fat mass and stores up nutrients to help it survive outside the womb.
  - Immune system and lungs develop to a point where the foetus can breathe without external support.
  - All organ systems mature in preparation for birth.
  - The baby is born approximately 266 days after coception or 280 days from the onset of the last period

# Behavioural capacities of the fetus

- Fetal Behavior • A fetus can swallow, move its arms and legs, yawn, and suck.
- The fetus gets ready to breathe outside the womb by “fetal breathing
- Touch: The fetus’s body parts come in contact with one another. A fetus sucks its thumb. It is the most developed sense.
- Taste: The fetus can detect the flavors of the amniotic sac. It likes sweets.

- 
- Smell: The fetus can detect the smells of the amniotic fluid, which are influenced by what the mother eats.
  - Hearing: Fetus responds to the sounds of the mother's voice, of others talking to the mother, and of the mother's blood pumping and her breathing.
  - Sight: The fetus may react to a bright light shown against the mother's skin, but its visual experience is very slight least developed.

- 
- Fetal Learning :Habituation—the decrease in response to repeated or continued stimulation—begins at about 32 weeks.
  - All these behavioural capacities enables the foetus to well prepared for independent living as a newborn baby.
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# FACTORS AFFECTING PRENATAL DEVELOPMENT

- Babies may be born with defects that range from gross anomalies that may lead to sudden death or to minimal physical or mental defects(**congenital defects**).
- These congenital defects could be caused by:
  - Abnormal genes and chromosomes
  - Harmful conditions in the womb( Environmental influences)
  - Abnormalities during the birth process(perinatal)

- 
- **Abnormal genes and chromosomes**
  - **Genetic abnormalities**
  - Result from mutations.-change in the chemical structure or arrangement of one or more genes that have the effect of producing a new phenotype



- **Phenylketonuria (PKU)**-genetic disorder involving a recessive gene where one can not effectively metabolize protein and so causes mental retardation and hyperactivity.
- **Hemophilia** –lack of the clotting factor hence excessive internal and external bleeding. Mostly in males.

- **Sickle-cell anaemia**-it's a genetic disorder affecting the red blood cells. A red blood cell is usually shaped like a disc but in sickle-cell anaemia, a change in a recessive gene modifies its shape to be hook-shaped.
- The cells die quickly hence causing anaemia and early death of the person due to failure to carry oxygen to the body cells through blood.

- 
- **Diabetes-** inherited condition in which the individual is unable to metabolize sugar properly because the body cannot produce insulin.
  - **Chromosomal abnormalities:**
    - Occurs during the process of cell division after conception either due to
    - Non separation of chromosomes

- 
- Deletion of chromosomes which may lead to loss of genes
  - Translocation of chromosomes ie deleting and attaching themselves to the wrong ones. The conditions include
  - **Down syndrome**-most common genetically transmitted form of mental retardation caused by the presence of an extra chromosome (47<sup>th</sup> ).

- Symptoms include round face/oriental appearance , square heads, protruding tongue, short and storky, defective hearts, retardation of mental and motor abilities. Its rare with Africans
- **Klinefelters syndrome**-sex chromosome abnormality where males have an extra X chromosome making them XXY instead of XY.

- Males with this have undeveloped testes, enlarged breasts, high-pitched voices, no facial hair after puberty etc.
- **Turner syndrome-** females without one X chromosome making them XO instead of XX. They are short in stature and have a webbed neck, mostly mentally retarded and sexually underdeveloped. Occurs once in every 3,000 in female births

- **The XYY syndrome /supermen-**is a genetic disorder where male have an extra Y chromosome. They are likely to be tall, more aggressive and violent , criminals etc



# Environmental influences (teratogens)

- Teratogen refers to any environmental agent that "can" harm a developing embryo (critical period) or fetus.
- It causes deviations in normal development and can lead to abnormalities such as
- physical deformities, severely retarded growth, blindness, brain damage or even death.
- They include:



# Mothers Nutrition/ Diet

- The growing fetus gets nourishment from maternal blood stream through placenta.
- The pregnant mother's diet should not only be enough but also balanced for normal prenatal development to occur.
- If not it may lead to retarded fetal growth, malformations, less developed brain, greater vulnerability to diseases, low birth weight, cerebral palsy and epilepsy

# Maternal Emotional state

- When the mother is happy and calm, it allows your **baby** to develop in a happy, calm environment.
- Extreme maternal stress may be related to lower birth weight and children with emotional problems and behavioral disorders
- Chronic feelings of worry and anxiety may yield to hyperactive and irritable infants.

# Mother's age

- Pregnancy most likely to result in a healthy baby if mother is in her 20's.
- Younger mother are at greater risk for inadequate diets and prenatal care and are more likely to have children with behavioral problem
- Older mothers(35+) have increased risk of miscarriage and stillbirth and are more likely to give birth to Down syndrome children.

# Maternal diseases

- **Exposure to certain viral, bacteriological and parasitic disorders can affect the baby's development e.g.**
- **Rubella** – German measles; causes blindness, deafness, heart abnormalities.
- **AIDS** –Infants born to HIV positive moms may acquire the disease; via prenatal barrier or during delivery.
- **Syphilis**- Miscarriage, serious eye, ear, bone and brain damage
- **Gonorrhea**- blindness

# Rhesus Incompatibility

- Condition in which an individual with Rh-negative blood develops antibodies after exposure to Rh-positive blood
- Incompatibility occurs when a mother is Rh-negative and the fetus is Rh-positive.
- The mother's blood forms antibodies which may destroy the red blood cells of the baby.
- harm the baby.

- 
- This can lead to either a miscarriage or a mentally retarded baby.
  - It doesn't affect the first baby.
  - A drug called Rhogam is usually given to prevent formation of antibodies.

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# Drugs-drugs and their known effects:

- **Thalidomide** was given for nausea to pregnant women; effects - no limbs, vision & hearing deficits.
- **Tobacco** – higher chance of stillborn and death at birth. Infants have lower birth weight.

# Teratogens cont...

- **Cocaine** – addictive stimulant; infants of cocaine using mothers are irritable, liable to react excessively to stimulation, uncoordinated, and slow learners (crack babies)
- **Methadone/heroin** – infants are born addicted to these and must receive it after birth to aid with withdrawal; these infants are premature, underweight, prone to respiratory illness, and have low attention spans.



# Alcohol

- Infants are born abnormal and/or have **fetal alcohol syndrome** whose characteristics are:
  - small head ( microcephaly)
  - underdeveloped brain,
  - congenital heart disease,
  - facial malformations,
  - and joint anomalies

# Teratogens cont...

- **Radiation** – It causes gene mutations in pregnancy it may lead to spontaneous abortion, mental retardation, microcephaly and leukaemia( blood cancer).
- **Paternal factors**-exposure to drugs, radiation, pesticides may result in abnormal sperm hence genetic abnormal children. Low vitamin C uptake also leads to children with birth defects and certain types of cancer. Fathers' advanced age contributes to about 5% of all cases of Down syndrome, dwarfism etc

# Perinatal- Birth Process

- Though born healthy the following may affect the baby at birth
- **Anoxia-** lack of sufficient oxygen due
  - to prolonged labour
  - Failure to breathe when separated from the mothers supply of blood.
  - Use of pain killers during birth
  - Too much anesthesia
  - This could lead to brain damage, death, cerebral palsy.

- **Use of instruments;** Forceps and vacuum extraction may be used if baby is too big or mother not ready. –not well developed to deliver
  - Blood vessels may break leading to brain damage, epilepsy, death, cerebral palsy.
- **Unhygienic conditions-** Baby may get infectionse.eg tetanus

Forceps



Vacuum extraction



# The neonate (Birth- 1 month)

- The Neonate is a newborn baby up to about four weeks
- Birth Averages
  - Average weight is 3.3 kgs
  - Average length is 51 cms (20 inches)
  - **Physical characteristics**
  - Looks abnormal, ugly
  - Wet, sticky, brown/black



- **Wrinkled skin**
- **Oval heads-misshapen heads ( unusual, undesirable shape but shape normalizes over time.)**
- **Flattened noses**
- **Eyes look swollen and bruised**

# Physical characteristics

- Skin coated with cheese-like protective substance called **vernix caseosa**
- Soft fine hair called **lanugo** that covered foetus during 5<sup>th</sup> and 6<sup>th</sup> month may be present and last a few weeks
- Hair is dark and soft though some born bald



- **6 soft spots called fontanelles on the head with the most visible being on top of the head. These harden/close by 1 ½ years.**
- **May have enlarged breast tissue, sometimes with discharge, due to maternal hormones.**
- **Some female neonates may experience a small amount of menstrual flow or discharge due to**

# Test used to ascertain normality

- To find out whether the neonate is normal and has no problem at birth the **Apgar score** is taken one minute and five minutes after birth
- The attendant observes the pulse, breathing, muscle tone, general reflex response and the colour of the skin( for blacks the palm or soles)
- A perfect Apgar score is 10 points; 7 or more is considered normal;

## Apgar scoring system in neonates

| Characteristics   | 0           | 1                              | 2                                |
|-------------------|-------------|--------------------------------|----------------------------------|
| Pulse/ Heart rate | Absent      | Less than 100 beats/min        | More than 100 beats /min         |
| Breathing         | Absent      | Slow/ irregular                | Strong cry                       |
| Muscle tone       | Limp        | Some flexion of extremities    | Active motion                    |
| Reflex responses  | No response | Grimace                        | Vigorous cry, sneezing coughing, |
| Skin colour       | Blue/ Pale  | Pink on body proximities/ blue | Completely pink                  |

# Behavioural capacities of neonate

- At birth, neonates have reflexive behaviours. These are automatic responses that occur involuntarily through some specific stimulus
- It has **survival reflexes** which help the baby to survive outside the womb. These are;
- e.g. breathing reflex, eye-blink, pupillary, rooting reflex, swallowing and sucking reflex

- **Primitive reflexes** which help assess the baby's developmental level and disappear with time.
- pointers to whether the neonate is normal or not. e.g babinski, palmer grasping, moro reflex, swimming reflex, stepping reflex
- These reflexes are essential for the infant's survival and are indicators of a healthy nervous system.

- **Moro Reflex (Startle Reflex):**
- **Description:** When the infant feels a sudden loss of support, or there is a loud noise, they will spread out their arms (abduction), then bring them back together (adduction) while crying.
- **Importance:** This reflex helps assess the development of the infant's nervous system. It typically disappears by 4-6 months of age.

- **Rooting Reflex:**
- **Description:** When the corner of the baby's mouth is touched, the infant turns their head towards the stimulus and opens their mouth, looking for the breast or bottle.
- **Importance:** This reflex is crucial for feeding as it helps the baby locate the source of food. It usually disappears around 4 months.





- **Babinski Reflex:**

- **Description:** When the sole of the foot is stroked from heel to toe, the big toe extends upward, and the other toes fan out.
- **Importance:** In infants, this is a normal response and typically disappears by 12-24 months. In older children and adults, it can indicate neurological problems.





- **Sucking Reflex**

- **Description:** When something touches the roof of the baby's mouth, such as a finger or a nipple, they will instinctively begin sucking.
- **Importance:** Essential for breastfeeding and bottle-feeding. This reflex is present at birth and becomes voluntary over time, generally by 2-4 months.

- **Palmar Grasp Reflex:**
- **Description:** When the baby's palm is stroked or touched, they will grasp the object tightly.
- **Importance:** This reflex helps assess neurological function and typically disappears by 3-4 months, at which point voluntary grasping begins.

- **Stepping (Walking) Reflex:**
- **Description:** When the baby is held upright with their feet touching a flat surface, they will make stepping movements as if they are trying to walk.
- **Importance:** This reflex is present at birth and fades around 2 months as voluntary walking skills develop later.

# Significance of Neonatal Reflexes

- Neonatal reflexes are important markers of neurological health and development.
- The presence, absence, or persistence of these reflexes beyond typical ages can help assess whether a newborn's nervous system is functioning correctly.
- For example, the persistence of the Moro reflex beyond 6 months or the Babinski reflex beyond 2 years may suggest underlying neurological problems.





## Five Infant Reflexes



a.



b.



c.



d.

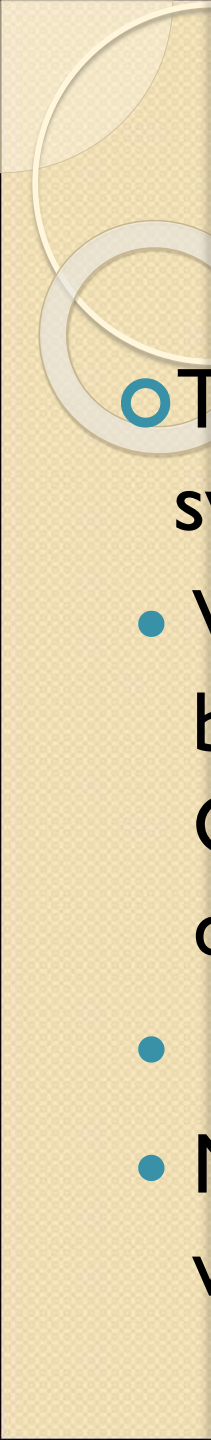


e.



# Sensory & motor skills development.

- **Newborns have all five senses.**
- **Hearing:** Newborns hear fairly well and respond most noticeably to high-pitched and loud sounds; the sound of mother's voice, etc
- **Smell:** has a strong sense of smell. how mother smells.
- **Touch:** Is highly developed as they respond to the reflexes. Especially developed around the mouth. It craves

- 
- Taste: Newborns recognize and prefer sweet tastes to sour, bitter, or salty ones.
  - Vision is developing quickly but is believed to be the weakest of the senses. Can recognize mother's face after a few days.
  - **Other skills**
  - Must establish clear regular sleeping and waking cycles

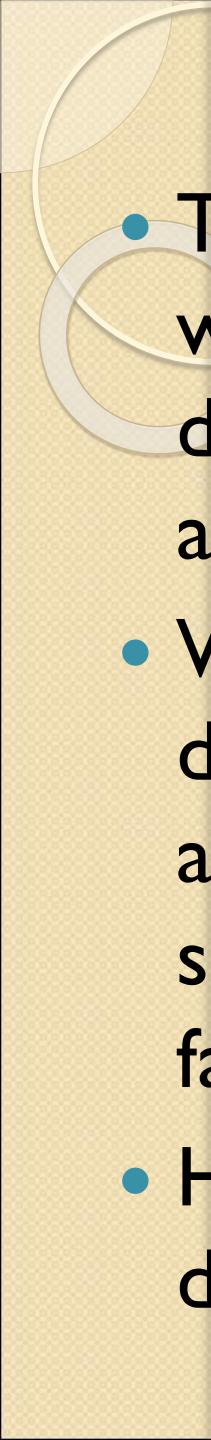


# Factors influencing neonatal development

- Baby's size at birth:
- Mother's health before conception, during pregnancy and after birth
- Environmental exposures
- How long the mother carried to term
- Nutrition
- The environment and culture of the parents

# HAVIGHURST'S DEVELOPMENTAL TASK THEORY

- Robert Havighurst emphasized that learning is basic and that it continues throughout life span.
- He argues that an individual moves from one stage to the next by means of successful resolution of problems or performance of developmental tasks.
- When people successfully accomplish and master these developmental tasks, they feel pride and satisfaction, and consequently earn the approval of their community or society.

- 
- This success provides a sound foundation which allows them to accomplish developmental tasks that they will encounter at later stages.
  - When people fail to accomplishing a developmental task, they're often unhappy and are not accorded the desired approval by society, thus experience of difficulty when faced with succeeding developmental tasks.
  - Havighurst states that growth and development occurs in six stages namely:

# **I Infancy and Early Childhood( Birth – 5yrs):**

1. Learning to walk.
2. Learning to take solid foods
3. Learning to talk
4. Learning to control the elimination of body wastes
5. Learning sex differences and sexual modesty
6. Forming concepts and learning language to describe social and physical reality.
7. Getting ready to read

## **II Middle Childhood ( 6-12yrs):**

1. Learning physical skills necessary for ordinary games.
2. Building wholesome attitudes toward oneself as a growing organism
3. Learning to get along with age-mates
4. Learning an appropriate masculine or feminine social role
5. Developing fundamental skills in reading, writing, and calculating



6. Developing concepts necessary for everyday living.

7. Developing conscience, morality, and a scale of values

8. Achieving personal independence

9. Developing attitudes toward social groups and institutions



### **III Developmental Tasks of Adolescence( 13-17yrs):**

1. Achieving new and more mature relations with age-mates of both sexes
2. Achieving a masculine or feminine social role
3. Accepting one's physique and using the body effectively
4. Achieving emotional independence of parents and other adults



5. Preparing for marriage and family life

6. Preparing for an economic career

7. Acquiring a set of values and an ethical system as a guide to behavior; developing an ideology

8. Desiring and achieving socially responsible behavior



## **IV Developmental Tasks of Early Adulthood( 18-35yrs):**

1. Selecting a mate
2. Achieving a masculine or feminine social role
3. Learning to live with a marriage partner
4. Starting a family
5. Rearing children
6. Managing a home
7. Getting started in an occupation
8. Taking on civic responsibility
9. Finding a congenial social group



## **V Developmental Tasks of Middle Adulthood(36-60yrs):**

1. Achieving adult civic and social responsibility
2. Establishing and maintaining an economic standard of living
3. Assisting teenage children to become responsible and happy adults
4. Developing adult leisure-time activities
5. Relating oneself to one's spouse as a person
6. Accepting and adjusting to the physiological changes of middle age
7. Adjusting to aging parents.

## **VI Developmental Tasks of Late Adulthood( over 60yrs):**

- 1. Adjusting to decreasing physical strength and health
- 2. Adjusting to retirement and reduced income
- 3. Adjusting to death of a spouse
- 4. Establishing an explicit affiliation with one's age group
- 5. Meeting social and civil obligations
- 6. Establishing satisfactory physical living arrangement.

# Physical and Motor Development from Infancy to late childhood

- **Infancy** (birth to about 2 years) is marked by rapid physical growth basic motor and sensory development (growth spurt),
- Childhood (early, middle and late) is the developmental period from about 2 to 12 years marked by steady physical growth,

# PHYSICAL AND MOTOR DEVELOPMENT FROM INFANCY TO 12 YEARS

## Physical Development.

- This involves changes in height and weight as well as development of the ability to carry out certain motor tasks
- The changes in weight and height are indicators of growth in childhood and adolescence
- A child's ability to perform certain physical tasks depends on the maturation of the neurons in the brain as well as maturation of the muscles and skeletal system.

- 
- The implication is that until a child reaches a state of readiness (maturation) no amount of training or practice will make them perform a task.

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# Physical growth

Weight : **the infant gains :**

- Birth to 4 months  $\rightarrow \frac{3}{4}$  kg /month
- 5 to 8 months  $\rightarrow \frac{1}{2}$  kg / month
- 9 to 12 months  $\rightarrow \frac{1}{4}$  kg /month

The infant will double his birth weight by 4-5 months and triple it by 10-12 months of age



# Height

- Length increases about 3 cm /month during the 1<sup>st</sup> to 3 months of age,
- then it increases 2 cm /month at age of 4-6 months,
- Then, at 7 – 12 months, it increases 1 ½ cm per month



# Motor development

- Motor development involves two tasks namely;

- **Locomotion**

- Body control and movement- It involves use of gross motor skills / using the large muscles in the body to make large movement. E.g. lifting head, sitting, standing with support, and alone, walking, kicking a ball etc.

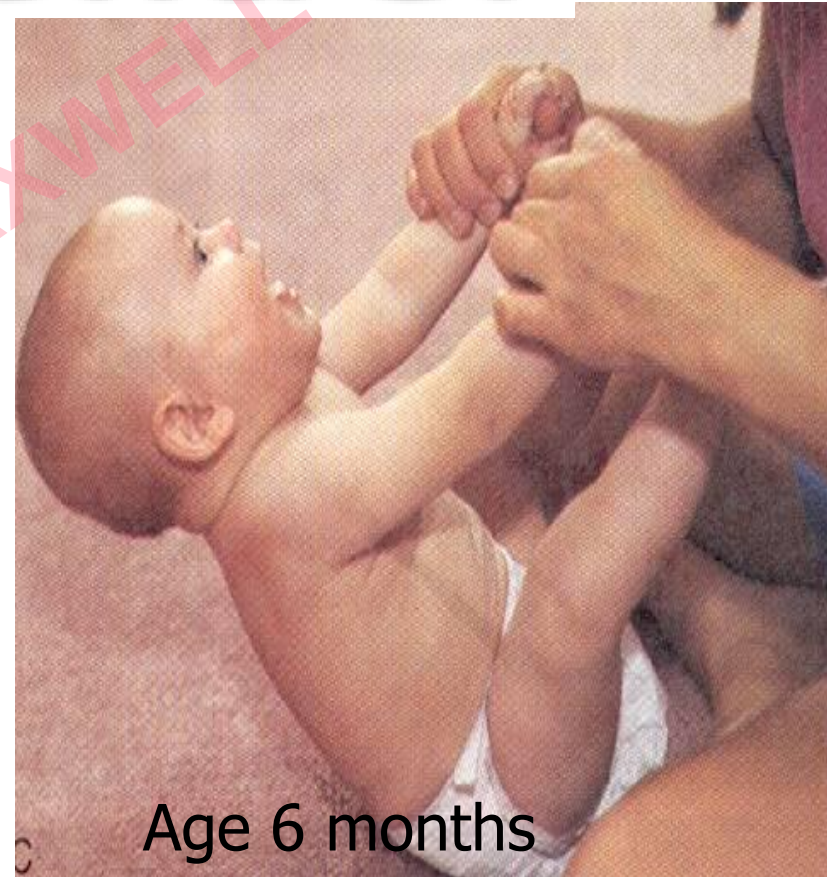
- **Prehension.**

- It is the use of fine motor skills i.e. engaging in the small muscles in the hands and fingers

- 
- These activities require eye-hand coordination
  - Includes, reaching out, holding, seizing, grabbing objects, grasping, picking objects, use of thumb and forefinger etc.
  - E.g. buttoning up a shirt, holding a pencil, picking up food, shaking musical instruments

# LOCOMOTION

## Head Control





# LOCOMOTION



# LOCOMOTION





# LOCOMOTION



# PREHENSION



# Motor Growth in Stages

- **At 2 months**
  - Hold head erect in mid-position.
  - Turn from side back.
- **At 3 months**, the infant can
  - Hold head erect and steady.
  - Open or close hand loosely.
  - Hold object put in hand



# At 4 months, the infant can:

- Sit with adequate support.
- Roll over from front to back.
- Hold head erect and steady while in sitting position.
- Bring hands together in midline and plays with fingers.
- Grasp objects with both hands.

# At 5 months, the infant can:

- Balance head well when sitting.
- Sit with slight support.
- Pull feet up to mouth when **supine**-lying on the back.
- Grasp objects with whole hand
- Hold one object while looking at another

# At 6 months, the infant can:

- Sit alone briefly.
- Turn completely over ( abdomen to abdomen ).
- Lift chest and upper abdomen when prone.
- Hold own bottle.

# At 7 months, the infant can:

- Sit alone.
- Hold cup.
- Imitate simple acts of others.

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# At 8 months, the infant can:

- Sit alone steadily.
- Drink from cup with assistance.
- Eat finger food that can be held in one hand.

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# At 9 months, the infant can:

- Rise to sitting position alone.
- Crawl (i.e., pull body while in prone position).
- Hold own bottle with good hand-mouth coordination

## At 10 months, the infant can:

- Creep well (use hands and legs).
- Walk but with help.
- Bring the hands together.

## At 11 months, the infant can:

- Walk holding on furniture.
- Stand erect with minimal support



# At 12 months , the infant can:

- Stand-alone for variable length of time.
- Sit down from standing position alone.
- Walk in few steps with help or alone (hands held at shoulder height for balance).
- Pick up small bits of food and transfers them to his mouth

- **13 month old:** walk and toddle( walk unsteadily) quickly
- **14-15 months old:** Most walk well, can run, walk upstairs with two feet on the same stair
- **18 months:** Most toddlers will attempt more complex skills. E.g. drinking from a cup independently, trying to dress themselves, using a crayon or pencil

- **22 months** : Walk in steps
- **24 months** ; Toddlers' fine motor skills become more sophisticated.
  - They may start to show an interest in scribbling, drawing, trying to write
  - Can jump over small objects, throw a small ball or object, walk up and down stairs, drinking from a cup independently, kick ball,

# Motor Development



## Six Motor Milestones (Infancy to 2 years)

- 1) Raising head and chest- 2-4 months
- 2) Rolling over- 2-5 months
- 3) Sitting up with support- 4-6 months
- 4) Sitting up without support- 6-7 months
- 5) Crawling- 7-8 months
- 6) Walking- 8-18 months

- **At three years-** the child can jump from bottom starting with two feet together, walk backwards, ride a tricycle
- **At 4 years-** swirl, swing, somersault, and descend short steps with alternating feet, dress themselves
- **5 years-** They are stronger, more coordinated, better balanced.



- 
- They perform motor tasks with greater skill and precision
  - Run skillfully, jump greater height with more skill, hop well on one foot, climb skillfully, skip alternating feet, jump rope
  - Copy letters, numbers,
  - Show a preference for one hand over the other



- **6 years**

- Enjoys physical activities, has improved balance.
- Arm and leg movement are well coordinated, gain speed and steadiness of arm movement.
- Good eye-hand coordination
- Performs complex skills like high jump, broad jumping, riding bicycle- hop, skip, jump



- **7 years** are graceful, speedy, agile, enjoy games which allow him to sit and rest
- Control of arms , shoulders improve, can write well, can walk on high narrow fence
- **8 years-** enjoy playing ball, rope jumping, swimming, dancing, climbing trees, organized games and sports
- **9-10 years-** demonstrate greater skills in gross motor activities- movement and coordination of whole body and well developed eye-hand coordination.
- By 12 years the child has perfected all basic motor skills

# Factors affecting physical and motor development

- **Genetics (Heredity):** Determines a child's potential for height, body build, muscle growth, and motor coordination.
- Inherited conditions or genetic disorders can influence overall physical abilities.
- **Nutrition** –sufficient, well balanced to prevent stunted growth, nutritional deficiencies such as kwashiorkor (lack of proteins, hence brown hair, swollen tummy), rickets ( vit D and calcium), Marasmus (lack of all types of food- starvation ), obesity

- 
- **Prenatal Factors:** Maternal nutrition, health, and exposure to toxins affect fetal growth.
  - **Premature birth** can lead to slower physical and motor skill development.
  - **Diseases-**Frequent illnesses, chronic diseases, and infections can interfere with steady growth
  - Healthy children gain strength, endurance, and coordination more easily

- 
- **Child rearing practices-** practice, stimulation, encouragement

- **Physical Activity and Practice**

- Opportunities for play, exercise, and exploration strengthen muscles and improve coordination.
- Lack of activity leads to weak muscles, poor balance, and delayed motor skills.

# Importance of Physical and motor development

- Stimulates intellectual growth and creativity
- Promotes good mental health
- Strengthens self-concept
- Encourages socialization, independence and self-confidence



COGNITIVE DEVELOPMENT



# COGNITIVE DEVELOPMENT

- **Cognition** is the mental activity through which human beings acquire and process information and knowledge so as to understand the world.
- **Cognitive development** refers to the changes that occur in children's mental skills and abilities over time such as attending, perceiving, learning, thinking and remembering



# Jean Piaget's Theory of Cognitive Development

- Jean Piaget proposed that children actively construct knowledge as they interact with their environment.
- He states that cognitive development occurs through the interaction of innate capacities and environmental events
- He believed that learning is a cycle of:
- **Assimilation** → using what you know
- **Accommodation** → adjusting your thinking
- **Equilibration** → reaching a balanced understanding

# ASSIMILATION AND ACCOMMODATION IN PIAGET'S THEORY



## 1. ASSIMILATION

New information is fitted into existing schemas



## 2. ACCOMMODATION

Schemas are modified to accommodate new information

## EQUILIBRATION



A balanced understanding is achieved

- **Assimilation:** Assimilation is the process of fitting new information into existing cognitive schemas (mental frameworks).
- The child uses what they already know to interpret new experiences
- **Accommodation:** Accommodation is the process of modifying existing schemas or creating entirely new ones when new information does not fit.
- When assimilation fails, the child adjusts their thinking.

- This leads to cognitive growth and more accurate understanding
- **Equilibration** enables children to move from one stage of cognitive development to the next.
- Piaget proposed that children's thinking develops in **four** stages.
  - i. **Sensorimotor**
  - ii. **Preoperational,**
  - iii. **Concrete operational**
  - iv. **Formal operational period**
- Each stage represents a qualitatively different way of understanding the world.

# Sensorimotor 0-2years

- The infants learn about the world through their **sensory experiences** and **motor actions**.
- Learning occurs through seeing, smelling, hearing, touching, and tasting and their actions (moving around and exploring its environment).
- During this stage the infant lives in the present.
- It does not yet have a mental picture of the world stored in its memory therefore it does not have a sense of object permanence
- <http://www.youtube.com/v/nwXd7WyyWNHY>



# Stages of sensori-motor

- Piaget divided the sensorimotor stage into **six progressive sub-stages**, showing how infants move from simple reflexes to intentional, symbolic thinking..

## *i) Simple reflexes Birth-6 weeks“*

- Coordination of sensation and action through reflexive behaviors
- These reflexes begin to become voluntary actions.
- For example, the palmer reflex becomes intentional grasping.

## *ii) Primary circular reactions (6 weeks-4 months)“*

- Infants repeat behaviors that are **accidental but pleasurable**.
- Actions focus on **their own body**.
- **Example:**  
A baby accidentally sucks their thumb and repeats it because it feels good.

### iii) **Secondary circular reactions(4-8 months)**

- Repetition of actions that produce interesting effects on the **environment**.
- Beginning awareness that they can cause events.
- **Example:**  
Shaking a rattle repeatedly to hear the sound.

### iv) **Coordination of secondary circular reactions** stages 8–12 months

- Coordination of vision and touch—hand-eye coordination; coordination of schemas and intentionality".<sup>1</sup>
- Beginning of **object permanence** (searching for hidden objects).



## v) *Tertiary circular reactions*, 12–18 months

- Infants become intrigued by the many properties of objects and by the many things they can make happen to objects;
- **Trial-and-error experimentation.**
- They experiment with new behavior".
- **Example:**  
Dropping a spoon from different heights to see how it falls.

## ***vi) Internalization of schemas | 8- 24 months***

- Development of symbolic thought and internal mental images.
- Beginning of deferred imitation (imitating an action later).
- Problem-solving without trial and error.
- **Example:**  
A toddler sees someone open a box and imitates the behavior later.

# Achievements of Sensori- motor stage

- **Object permanence**
- Object permanence - Refers to *the child's ability to know that an object exists even when the object cannot be seen or heard* them /even if it is hidden. It requires the ability to form a mental representation (i.e., a schema) of the object..
- **Development of imitation**
- Voluntary imitation is precise at 12-18 months
- **Deferred imitation**: The ability to produce the behavior of an absent model

# PRE- OPERATIONAL STAGE (2-7YEARS)

- Children acquire the ability to internally represent the world/ objects through words and images.
- They are able to use mental representations and language to describe, remember, and reason about the world.
- They also start to see the world from other people's viewpoint
- Evidence: **deferred imitation**- ability to imitate someone who is not present.
- **Imaginary play**: Using one object to stand for another.

# Limitations of preoperational Stage

- **Animistic thinking-** Believing that inanimate things are alive and have feelings and intentions.
- **Egocentric thought-** *unable to readily see and understand other people's point of view*
- **Lack of conservation-** the child believes that a given quantity of matter changes when a substance is rearranged or reshaped even though added or taken away.
- During a conservation test, the child thinks a taller glass has “more water” even if the amount is the same.





Two glasses with an equal amount of liquid

Source:

[http://en.wikipedia.org/wiki/Piaget%27s\\_concepts\\_of\\_conservation#mediaviewer/File:Conservation1.jpeg](http://en.wikipedia.org/wiki/Piaget%27s_concepts_of_conservation#mediaviewer/File:Conservation1.jpeg)



The liquid from one of the short glasses has been poured into a taller, skinnier glass. A child that cannot conserve would assume the taller glass has more liquid than the shorter glass

Source:

[http://en.wikipedia.org/wiki/Piaget%27s\\_concepts\\_of\\_conservation#mediaviewer/File:Conservation2.jpeg](http://en.wikipedia.org/wiki/Piaget%27s_concepts_of_conservation#mediaviewer/File:Conservation2.jpeg)

# Limitations of preoperational Operational Stage

- **Centration-** tendency to focus on only one dimension of a stimulus and ignore other dimensions.
- **Irreversibility-** the child does not realize that any change in shape, position, or order of matter can be undone or reversed mentally.
- **Inability to reason about transformations-** the child focuses on static situations, they have difficulty thinking about change processes



# Limitations of preoperational Stage

- **Single classification-** child can only classify objects in one way at a given time. E.g. A child denies that a mother can also be a doctor.
- **Transductive reasoning-** reasoning by combining unrelated facts simply because two events occur together. E.g. the child believes that the clouds cause the moon to grow.

# Concrete Operational Stage( 7-11 yrs)

- The stage is called concrete because children can think logically much more successfully.
- However, their thinking is still concrete and tied to tangible, real-world objects (concrete) and experiences or imaginable materials or pictures of them.
- During this stage, however, most children still can't think abstractly or hypothetically.

# Achievements of Concrete Operational stage

- They are able to understand the **principle of conservation**, the understanding that something stays the same in quantity even though its appearance changes.
- Can only think logically in concrete situations.
- **Reversibility**- the child understands that certain processes can be reversed. E.g. Understanding that subtraction is the reverse of addition.

# Achievements of Concrete Operational stage

- **Ability to reason about transformations-** the child can reason about change and its effects. E.g. Can understand the metamorphosis of a butterfly.
- **Multiple classification-** believe that objects may belong to several categories simultaneously. E.g. Acknowledging that a mother can also be a teacher, spouse, artist. Etc.
- **Deductive reasoning-** students can draw a logical inference from two or more pieces of information. E.g. A student deduces that if all children are human beings, and if human beings are living things, then all children must be living things.

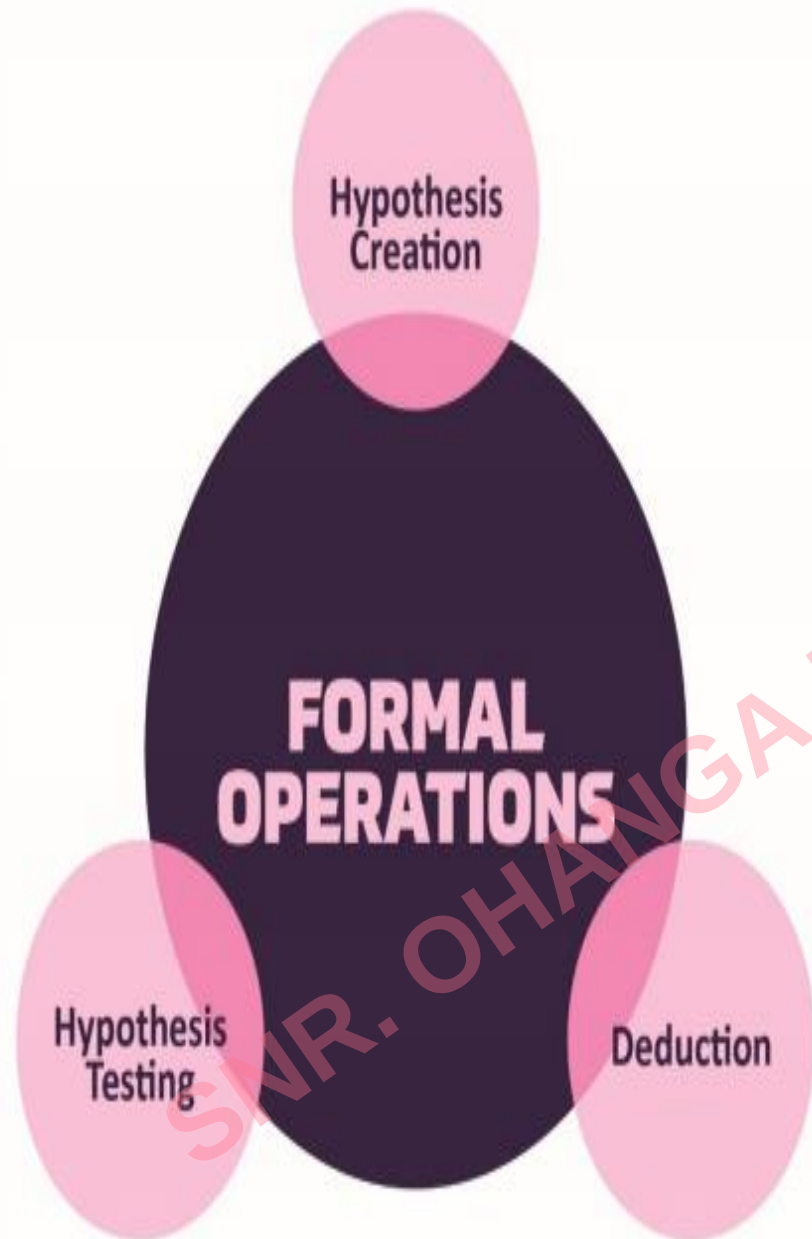


# COGNITIVE DEVELOPMENT IN ADOLESCENCE



# Formal Operational Stage II+

- Adolescents in this stage develop the ability to think abstractly and hypothetically.
- Thinking becomes much more sophisticated and advanced
- They can engage in logical reasoning and problem-solving that goes beyond concrete experiences.
- They are able to think about all the possibilities in a situation beforehand, and then test them systematically, because they are able to engage in true scientific thinking.
- They become capable of imagining situations that “might be,” rather than just those that actually exist.
- They can understand ethics & can think about possibilities and abstract concepts like justice, love, and freedom



Teenage thinking is characterized by the ability to reason logically and solve hypothetical problems



# Characteristics of Adolescents Thinking

- Once adolescents can understand abstract thoughts, they enter a world of hypothetical possibilities and demonstrate **egocentrism**, a heightened self-focus.
- The egocentricity comes from attributing unlimited power to their own thoughts.
- Piaget believed it was not until adolescents took on adult roles that they would be able to learn the limits to their own thoughts.

# Elkinds Theory of Adolescent Egocentrism

- David Elkind (1967) expanded on the concept of Piaget's adolescent egocentricity.
- Elkind theorized that the physiological changes that occur during adolescence result in adolescents being primarily concerned with themselves.
- Additionally, since adolescents fail to differentiate between what others are thinking and their own thoughts, they believe that others are just as fascinated with their behavior and appearance. thinking is often characterized by adolescent egocentrism.

# Forms of adolescent egocentrism

- **Imaginary audience-** is the adolescent's belief that those around them are as concerned and focused on their appearance as they themselves are.
- Belief that those around are always focusing on their behaviour and that *everyone is constantly watching them* 'All eyes on me'



"they must be whispering about me."

- 
- **Personal fable-** they regard themselves and their feelings as being special and unique.
  - E.g. Nobody else has ever experienced a particular emotional experience as they have, “you don’t know how it feels to be in love”
  - Adolescent belief that normal rules do not apply to them.



- **Naïve idealism-** the inability of the adolescent to differentiate between his own idealistic thoughts and the real world. The criteria for making judgement is what is logical and not what is realistic.
- **Invincibility Fable.** They also believe that they have a personal mission in life which makes them special, omnipotent, invincible, and therefore can take risks.
  - Thus are somehow immune to laws of mortality and probability.
  - Belief they are magically protected from harm
  - Adolescents will engage in risky behaviors, such as drinking and driving or unprotected sex, and feel they will not suffer any negative consequences.

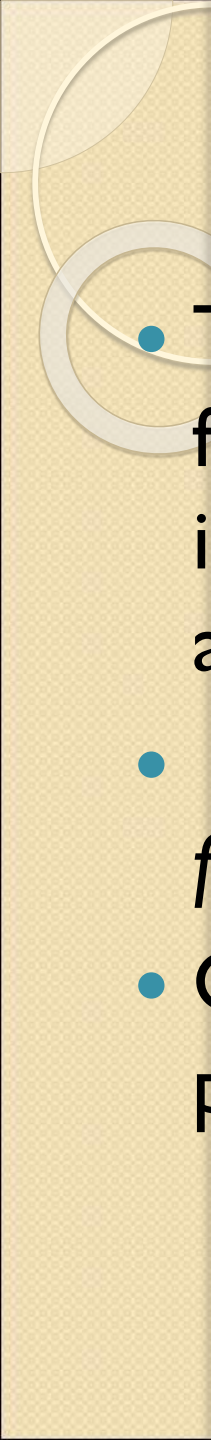
# Educational Implications

- The teachers should encourage the following within the classroom
- Focus on the process of learning, rather than the end product of it.
- Using active methods that require rediscovering or reconstructing "truths."
- Using collaborative, as well as individual activities (so children can learn from each other).
- Devising situations that present useful problems, and create disequilibrium in the child.
- Evaluate the level of the child's development so that a suitable tasks can be set.



# Vygotsky's Sociocultural Theory of Cognitive Development

- This theory is based on the following concepts and principles
  - Elementary mental functions ( EMF)
  - More Knowledgeable Other (MKO)-
  - Zone of Proximal Development (ZPD).
  - Role of Language
  - Scaffolding

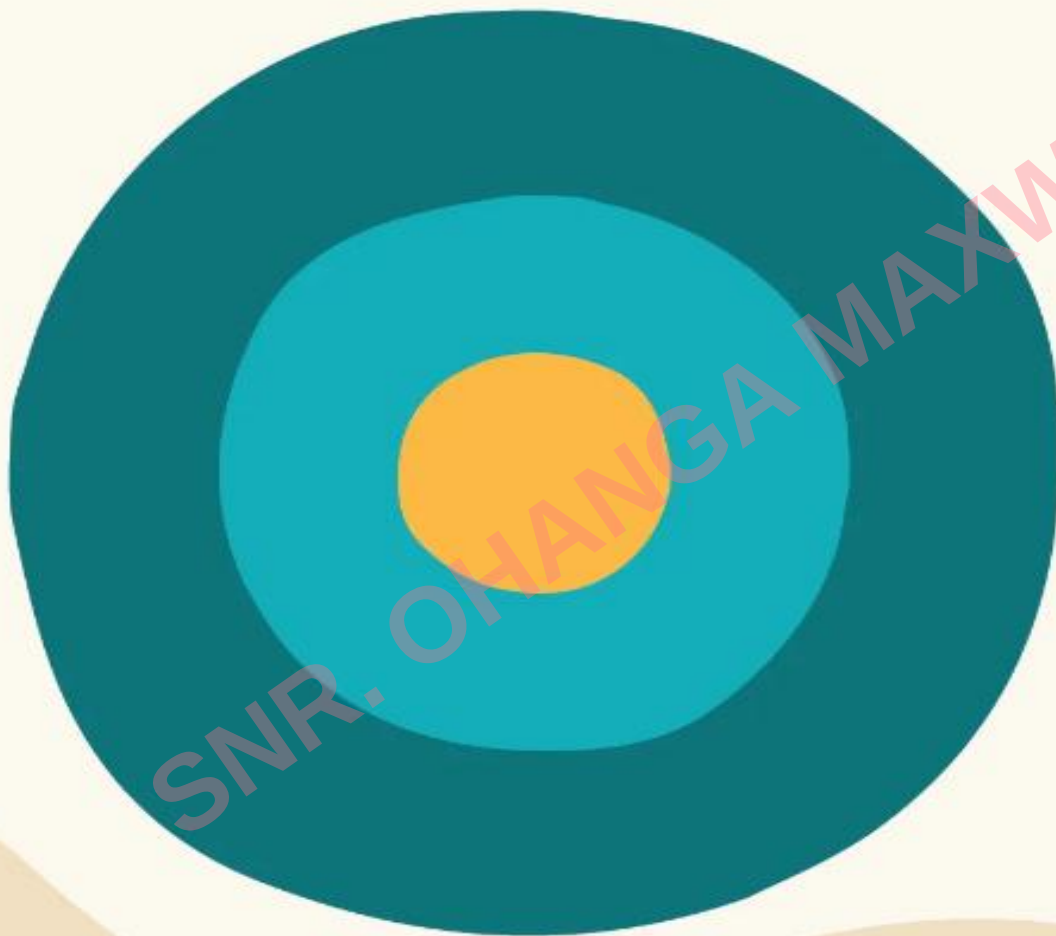
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- The Vygotsky's theory stresses the fundamental role of social and cultural interaction in the development of cognitive abilities.
  - He argues that cognitive development occurs *first on a social level*, then within the individual.
  - Children learn through interactions with parents, teachers, and peers.

- Vygotsky outlined three main concepts related to cognitive development:
  - (i) culture is significant in learning,
  - (ii) Language is root of culture and the most important cultural tool since it guides thinking, planning, and problem-solving.
  - (iii) individuals learn and develop within their role in the community.
- Culture can be defined as the morals, values, and beliefs of its community members, which are held in place with systems and establishments
- **Teaching relevance:** Use culturally relevant examples and tools to support learning.

# Zone of proximal development:

- It refers to the distance between what a learner can do unsupported (**independently**) and what they can do supported
- **(with guidance)**. The MKO (more knowledgeable other) can offer temporary support (scaffolding) to the child during the learning process.
- Provide tasks that are *challenging but achievable with support*.
- The ZPD is affected by the unique attributes of each individual, including personality, self-regulation, and previous knowledge.

# ZONE OF PROXIMAL DEVELOPMENT



● Tasks that learner cannot do even with assistance

● Tasks that learner can do with assistance

● Tasks that learner can do without assistance





The ZPD can be broken into three stages. Think of them as a series of overlapping circles:

1. **Tasks the learner can do without assistance.** This category includes everything a person can do without help from a more experienced individual.
2. **Tasks the learner can do with assistance.** This category includes tasks a person can't work through by themselves but can work through with help, also known as their ZPD.
3. **Tasks the learner can't do with assistance.** The final category includes tasks that are too difficult to perform even with an instructor's help. For example, a young child might be able to spell out their own name by themselves but might need help from someone else to write the complete alphabet. The task is above their skill level and outside their ZPD.







# The More Knowledgeable Other (MKO)

- In Vygotsky's framework, the “more knowledgeable other” is a term for someone who guides a learner through a new skill.
- This is anyone with a higher level of expertise (teacher, parent, peer, even a digital tool).
- Helps the learner progress through the ZPD.
- As the child gains more familiarity with the task and becomes more skilled at it, the MKO may then provide less guidance.
- **Teaching relevance:** Use peer teaching, teacher modelling and guided practice

# Concept of Scaffolding

- Scaffolding is a process that supports children as they learn to perform a task independently.
- Vygotsky's **concept of scaffolding** refers to the temporary support given to a child by a MKO to help a learner master a task.
- Support is gradually withdrawn as competence increases.
- **Teaching relevance:** Step-by-step guidance, modelling, cues, hints, questioning.





# Bruner's Theory of Cognitive development

- He emphasized that **learning is an active, constructive process.**
- Learners do not simply absorb information passively.
- They actively **construct meaning** by connecting new information to what they already know.



# Modes of Representation

- Bruner believed that children represent knowledge in three ways:
  - **a) Enactive 0-1 years- (Action-Based)**
    - Learning through physical actions.
    - Example: manipulating objects, performing tasks.
  - **b) Iconic 1-6 years (Image-Based)**
    - Learning through pictures, diagrams, and mental images.
  - **c) Symbolic 7 years onwards (Language-Based)**
    - Learning through symbols, language, formulas, and abstract thinking.
- These modes develop in sequence but **remain available throughout life**

# Cognitive Representations

## Jerome Bruner

### Learning Modes

#### Enactive

Learn through movement or action



Play with a book



#### Iconic

Learn through images or icons



Look at pictures



#### Symbolic

Learn through abstract symbols



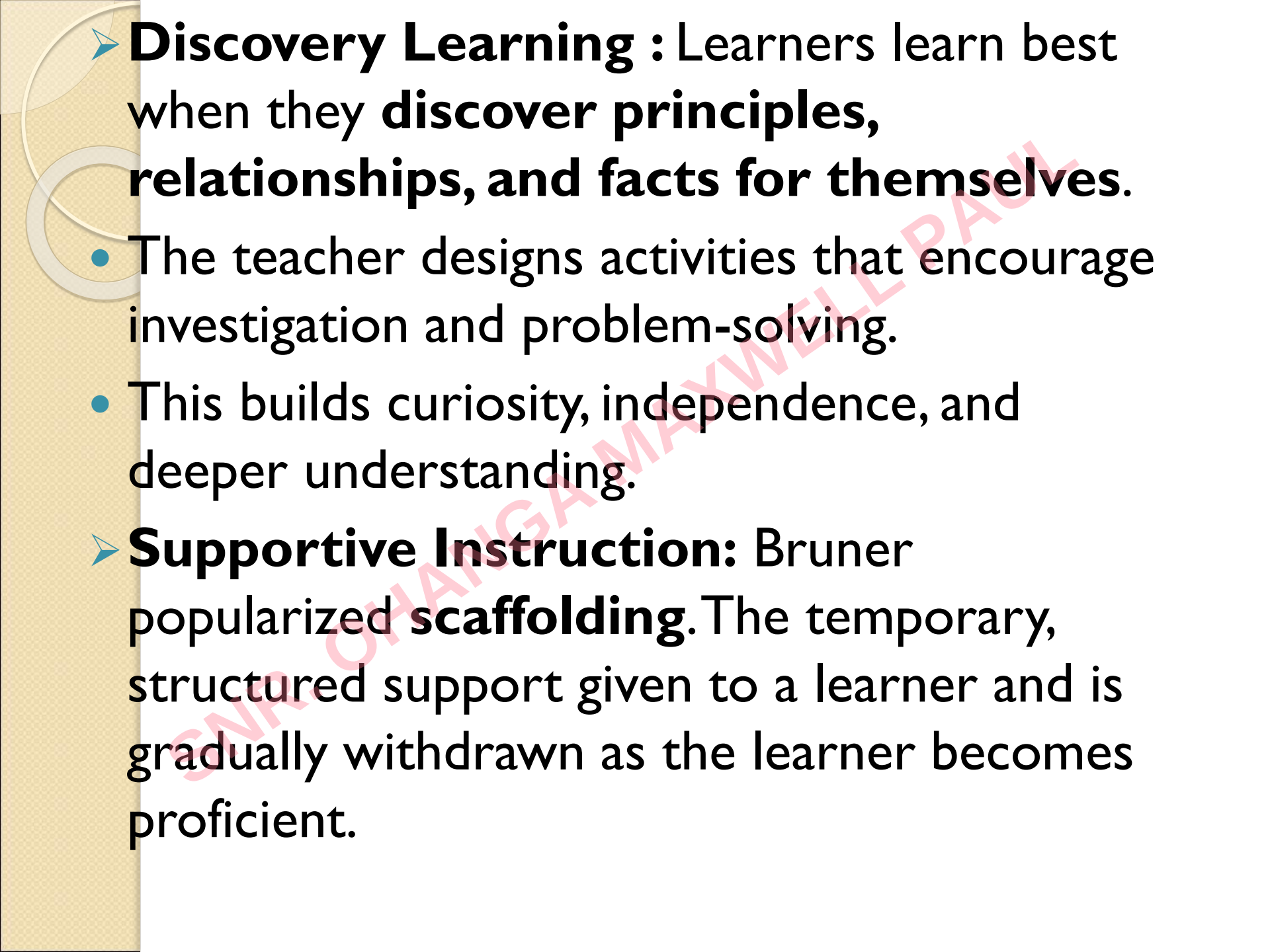
Read for research





# Spiral Curriculum

- Spiral curriculum refers to a curriculum design in which key concepts are presented repeatedly throughout the curriculum, but with deepening layers of complexity, or in different applications.
- **Bruner emphasized that:**
- Any subject can be taught at any stage of development if presented appropriately.
- Content should be revisited at increasing levels of complexity.
- Example:  
Fractions taught simply in early grades → more complex operations in upper grades

- 
- **Discovery Learning :** Learners learn best when they **discover principles, relationships, and facts for themselves.**
  - The teacher designs activities that encourage investigation and problem-solving.
  - This builds curiosity, independence, and deeper understanding.
  - **Supportive Instruction:** Bruner popularized **scaffolding**. The temporary, structured support given to a learner and is gradually withdrawn as the learner becomes proficient.

- 
- **Intrinsic motivation** Bruner also emphasized learning driven by interest and curiosity.
  - **Knowledge Organization:** He stated that understanding how a student represents knowledge—through actions, images, or symbols allows educators to present the same content in multiple, accessible formats to accommodate diverse learning styles and foster deeper comprehension.

# Implications on the learning process

- Guides lesson planning from concrete to abstract.
- Teachers should assist learners in building their knowledge.
- Encourages active, student-centered learning.
- Supports individualized instruction through scaffolding.
- Promotes deep understanding through discovery and inquiry-based teaching.
- Helps teachers design curricula that build progressively (spiral curriculum)

# LANGUAGE DEVELOPMENT

- Language development is the process by which children acquire, understand, and begin to use language.
- It involves the progressive growth of linguistic skills, including the ability to listen, understand spoken and written words, and produce meaningful speech or written text.
- This process unfolds over several years, beginning in infancy and continuing through early childhood.
- It is influenced by a combination of biological, cognitive, social, and environmental factors.

# Theoretical views on language acquisition

- Several key theories explain language development, each emphasizing different aspects of how children acquire language skills. **Here are the main theories:**
- **Behaviorist Theory** - B.F. Skinner
  - He argues that language development is based on operant conditioning.
  - Children learn language through imitation, reinforcement, and shaping by caregivers.



# Nativist Theory

- **Noam Chomsky Theory**
- Chomsky's nativist theory argues that humans are biologically programmed to learn language.
- He proposed that all humans have an innate "Language Acquisition Device" (LAD) – a mental structure enabling them to acquire language and understand its rules.
- The idea that all languages share a common underlying structure that humans are innately equipped to understand.



- 
- **Social interaction Theory,**
  - Argue children learn language in the interactive and communicative context.
  - Emphasis is on social skills and effects of social environment
  - **Cognitive theory**
  - believe that children have the ability to categorize their world before they can communicate with others
  - They use language structures when their cognitive abilities enable them to do so

# Motherese

- **Motherese or child-directed speech (CDS)** plays a critical role in facilitating language development in children.
- It is a specialized form of speech that caregivers use when talking to babies and young children.
- Baby talk characterized by the shortening and simplifying of grammar and exaggerated speech melody



# **Stages of Language Development**

SNR. OHANGA MAXWELL PAUL

# **The pre-linguistic stage/ pre-speech ( birth- 12 months)**

- **Developmental aspects related to prespeech would include the**
- **Crying**
- **Cooing- repeatedly producing single vowel sounds begins e.g. 'ah' 'oh'.**
- ***Babbling*- repeating meaningless sounds and syllables that are the building blocks for later language development. E.g. gagaga, dadadada**
- **By 8 months:- Gestures are used to communicate about objects**



# Holophrase or one-word sentence (10-18 month) Stage

- Children begin to utter a single word at a time to express a whole sentence
- Its meaning is supplemented by the intonation and the gestures or non-verbal.
- An example of such a one-word sentence , child saying “mama” could mean “where is mama” or “there is mama” Others are doggie, dada etc
- The language is said to be **over- extended** which could be due to limited vocabulary



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# 15 MOST COMMON 1<sup>ST</sup> WORDS FROM BABIES





# Telegraphic speech( 18- 30 months)

## Stage

- It is the last stage of language before a child can speak fluently.
- Children put two words together e.g. baby chair, cat sleep?, Where ball, Not egg”  
“More sugar!”
- Sentences contain only the critical content words- like a telegram leaving out the articles.
- They progress and develop language at a much faster rate now that they have grasped the very essentials of language

# Telegraphic speech

refers to the two-word stage of language development.

Mommy  
see

Baby  
Draw



# **Acquisition of complex rules (3 yrs and above)**

- **Children pick up grammatical rules intuitively and apply them rigidly.**
- **Overregularization-** the act of inappropriately applying the grammatical rules for forming plurals and past tenses to irregular nouns and verbs.
- **They overgeneralize rules in ways that are inconsistent with common usage;**
- **for example, "I comed home" for "I came home", "I maked" for I made (sometimes called creative grammar).**
- **Correct forms are temporarily replaced as rules are internalized**

# Factors Influencing Language Development

- Social environment
- Cognitive development
- Maturation and other developmental factors e.g hearing
- Bilinguals: **Bilingual** means involving or using two languages



# WHAT IS PERSONALITY?



**Personality** refers to the unique set of characteristics, traits, behaviors, and patterns of thinking that define an individual's way of being in the world.

It encompasses various aspects of an individual's psychological makeup, such as temperament, values, beliefs, and emotions, and is shaped by a combination of genetic, environmental, and cultural factors.

**Personality** influences how people perceive and interact with others and can impact their life choices, relationships, and overall well-being. It can also change and develop over time through experiences and personal growth.

- Personality is an individual's characteristic or unique pattern of thinking, feeling, & acting/behaving.

- “Thinking”: Personality includes differences between people in how they typically think.

**Example:**

- Do people tend to focus on the positive (optimists) or the negative (pessimists)?

- “Feeling”: Personality includes differences between people in how they typically feel.

**Examples:**

- Do people tend to be happy or unhappy?
- Do they experience intense emotions or not?
- Do they get angry easily?
- Are they especially sensitive to rejection



- 
- “Behaving”: Personality includes differences between people in how they typically behave. **Examples:**
  - Do they tend to talk a lot?
  - Do they usually go along with what other people want or insist on doing things their way?
  - Are they neat and tidy or sloppy and disorderly?

**TABLE 1.1** Adjective check list

Make a check mark next to the words you believe apply to your personality.

|                    |                  |                   |
|--------------------|------------------|-------------------|
| _____ affectionate | _____ ambitious  | _____ assertive   |
| _____ boastful     | _____ cheerful   | _____ cynical     |
| _____ demanding    | _____ dominant   | _____ fearful     |
| _____ forceful     | _____ generous   | _____ high-strung |
| _____ impatient    | _____ insightful | _____ meek        |
| _____ moody        | _____ optimistic | _____ opinionated |
| _____ persistent   | _____ prudish    | _____ relaxed     |
| _____ sarcastic    | _____ sensitive  | _____ sociable    |
| _____ submissive   | _____ tolerant   | _____ trusting    |
| _____ uninhibited  | _____ vindictive | _____ withdrawn   |

# Determinants of Personality

- Personality is shaped by a combination of genetic and environmental factors
- **Genetic Determinants ( nature):**
  - These are the hereditary determinants of personality
  - The genes inherited from both parents lay down the potentials towards the development of various physical and psychological characteristics of an individual.

- 
- **Environmental determinants**
  - It refers to all the experiences that the person encounters all the way from the conditions of the uterus, the conditions surrounding birth, the early environments encountered in the home including the child-rearing practices the parents use.
  - The child's relationship with siblings, peers and friends too has power to influence personality one way or the other and the child's school experiences



# **THEORIES OF PERSONALITY DEVELOPMENT**

SNR. OHANGA MAXWELL PAUL

# PSYCHO-ANALYTIC THEORY BY SIGMUND FREUD

- **According to Freud, Personality involves several factors:**
- **Instinctual drives**( biological, innate )– food, sex, aggression- libido and thanato drives
- **Unconscious processes:** Personality characteristics are mostly a reflection of the contents of the unconscious mind.
- **Early childhood influences** (i.e. psychosexual stages) on adult behavior.
- Things that happen during childhood can be repressed and later affect behavior and mood e.g. child abuse, sibling rivalry etc



# PERSONALITY STRUCTURE

- According to Freud the personality is structured into three parts (i.e. tripartite)
- id
- ego
- superego
- all developing at different stages in our lives.

# Freud's view of personality



# Id

- The id is usually present from birth.
- Id consists of all inherited (biological) components of personality.
- It resides in the unconscious mind and works to satisfy basic urges, needs
- It **seeks immediate gratification/satisfaction.**
- It operates **on pleasure principle** and contains primitive drives/urges and needs.
- A person that is dominated by id becomes impulsive and narcissistic

# Ego

- This is the rational part of our personality.
- It is the part of us that is seen by others.
- It is the **decision making component** of personality
- The ego balances the pleasure seeking /aggressive drives of the Id with moral control of the superego in the context of reality.
- Ego operates on the **reality principle** and helps Id satisfy desires in a realistic way.

# Superego

- The **superego**
- It develops as a child interacts with others, learning the social rules for right and wrong.
- The superego acts as our conscience; it is called the ***moral arm of personality***.
- It operates to regulate the wishes of the id in order to ensure that the ego fulfils them in line with the acceptable code of morality
- It strives for perfection and judges our behavior, leading to feelings of pride or—when we fall short of the ideal—feelings of guilt.



# Interaction of Id, Ego and Superego

- Id and superego are always in constant conflict; id, seeking instant gratification does not care about the consequences while superego on the other hand tells us that we must behave in a socially acceptable way.
- Ego, the rational part, helps satisfy pleasure seeking desires in a rational way that will not lead to guilt.
- A person with a strong ego balances the two thus develops a healthy personality.
- If ego is not able to mediate the conflict between the id and superego, feelings of anxiety or **neurosis**— mental disorder marked by anxiety and fear result.
- A tendency to experience negative emotions result.
- Ego seeks to restore balance through protective measures called **defense mechanisms**.



# Defense Mechanisms

Defense Mechanisms – certain specific means by which the ego unconsciously protects itself against unpleasant impulses or circumstances.

- Rationalization
- Repression
- Denial
- Projection
- Reaction Formation
- Regression
- Displacement
- Sublimation



"My therapist says it's a defense mechanism."

# Defense Mechanisms

- **Repression:** Motivated forgetting: What you forget can't hurt you.
- **Projection:** In projection, you take what you think are unacceptable impulses and literally “project” (or attribute) them to others. If I hate you I say you hate me

- 
- **Displacement** Transferring your unacceptable feelings toward someone you're supposed to love (or perhaps fear) onto a safer target is the hallmark of displacement.
  - The classic example is that you're treated badly by your boss, so you go home and express your anger by yelling at someone in your family.

- 
- **Rationalization:** In this defense mechanism, you use an excuse to justify an experience that reflects negatively on you
  - **Reaction Formation:** you turn your unacceptable impulses into their opposite. If one believes drinking is bad and has desires for drinking the person unconsciously adapts hatred for alcohol

- **Sublimation:** It is a mechanism that causes channelization of socially unacceptable desires into acceptable form.
- A young man who has lost his lover may turn to write poetry about love.
- A person who has aggressive feeling and cannot express in society can become a soldier or boxer
- **Intellectualization:** Like rationalization, in intellectualization you come up with a reason to explain away the negative results of an event or encounter. You break your favorite butter dish.
- However, instead of allowing yourself to become even temporarily annoyed, you distance yourself entirely and just go on to put the butter in something else



- 
- **Denial:** Classic denial means that you deny having negative or harmful impulses; if you did, you'd be overwhelmed with anxiety.
  - **Regression:** Reverting to an earlier psychosexual stage when you felt happier and more secure (think of a thumb-sucking preschooler). You begin to act like a child.



# Psychosexual Stages of personality development

- Freud believed that the nature of the conflicts among the id, ego, and superego change over time as a person grows from child to adult.
- He maintained that these conflicts progress through a series of five basic stages, each with a **different focus: oral, anal, phallic, latency, and genital.**

- 
- Across these five stages, the child is presented with different conflicts between their biological drives (id)
  - and their social and moral conscience (superego) because their biological pleasure-seeking urges focus on different areas of the body (erogenous zones).

# Stage I: Oral stage (0-2 years)

- In the first stage of development, the pleasure-seeking urges is centered in a baby's **mouth**.
- During this stage, the baby gets pleasure from putting all sorts of things in its mouth to satisfy the libido thus satisfying the desires of the id.
- The desires of the id at this stage are oral such as sucking, biting or breastfeeding.
- A person who becomes fixated gets an oral personality. Such personalities include;
  - Nail biters
  - Finger chewers
  - Smokers
  - Thumb suckers
- The individuals tend to engage in such oral behaviors particularly when under stress.

# Psychosexual stages

- Fixation in this stage can lead to adult habits such as thumb-sucking, smoking, drink alcohol, over-eating, and nail-biting.
- Personality traits can also develop during adulthood that is linked to oral fixation; these traits can include optimism and independence or pessimism, gullible(easily persuaded), aggressive and hostile

# Psychosexual Stages

- **Anal Stage (18 months to three years)..-** The pleasure-seeking urges is centered the anus (erogenous zone)
- The child's focus of pleasure in this stage is on eliminating and retaining faeces.
- Parents stress toilet training and bowel control during this time period
- Smooth toilet training leads to development of self-confidence, creativity and productivity.
- Fixation in the anal stage can lead to anal-retention or anal-expulsion.
- If a child learns to retain faeces he/she develops **anal retention character**, be stingy, aggressive, excessively stubborn, obsession with cleanliness, perfection, and control.
- **Anal expulsive** leads to disorganized, destructive and messy characters.

# Phallic Stage ( 3 – 6 years).

- The pleasure-seeking urges ( erogenous zone) switches to the genitals.
- Freud believed that during this stage boy develops **unconscious** sexual desires for their mother. This results to **oedipus complex / conflict**
- Because of this, he becomes rivals with his father and sees him as competition for the mother's affection.
- During this time, boy also develops fear that the father will punish him for these feelings, such as by castrating him and thus develops **castration anxiety**



- Similarly, girls develop **unconscious** sexual attraction( love and desire) to their father. This conflict is termed the ***Electra Complex***.
- As a result they feel resentful they do not have a penis and they develop ***penis envy***
- To resolve this the boys vicariously possess the mother and the girls the father by identifying with the parents of the same sex.
- This process of identification enables them to adopt the appropriate sex roles and behavior
- Fixation at this stage may bring about sexual deviancy or weak sexual identity.

# Latency Stage ( 6-11 years).

- During this stage, libido is dormant. There is no pleasure-seeking sexual urges.
- No psychosexual development takes place and thus the name latent.
- Freud thought that sexual energy at this stage was **repressed** and **sublimated or channeled** toward school work, friendship and hobbies.
- The child's strives to master new skills and knowledge.
- Freud asserted that play is confined to children of the same gender.

# Genital Stage (puberty on).

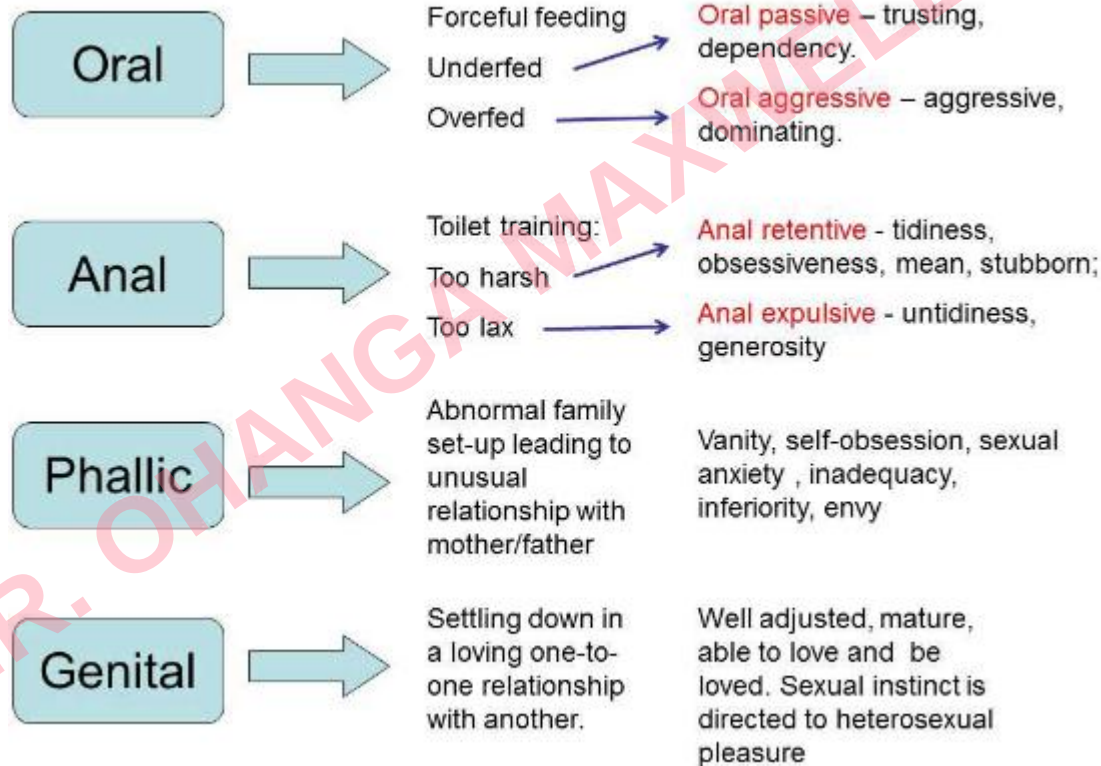
- The final stage of psychosexual development begins at the start of puberty when sexual urges are once again awakened.
- The adolescents direct their sexual urges onto opposite sex peers ( heterosexual relationship)
- The primary focus of pleasure is the genitals.
- They explore pleasure through more mature love and affection
- If one does not experience fixation in any of the psychosexual stages, he or she will grow into a well-balanced human being.

- Successful resolution of the stages leads to a healthy adult personality
- Freud suggested that some individuals, whose needs in a particular stage have not been met adequately develops **frustration**.
- On the other hand, if the needs are met way too well (**overindulgence**), the person according to Freud may be reluctant to leave the stage.
- Frustration or overindulgence result to fixation where individuals remain at a particular stage even as adults thus leading to **unhealthy personality traits**.

# Psychosexual Stages

|         |   | Location of the libido (source of pleasure) |
|---------|---|---------------------------------------------|
| Oral    | → | The mouth – sucking, swallowing etc.        |
| Anal    | → | The anus – withholding or expelling faeces  |
| Phallic | → | The penis or clitoris - masturbation        |
| Latent  | → | Little or no sexual motivation present      |
| Genital | → | The penis or vagina – sexual intercourse    |

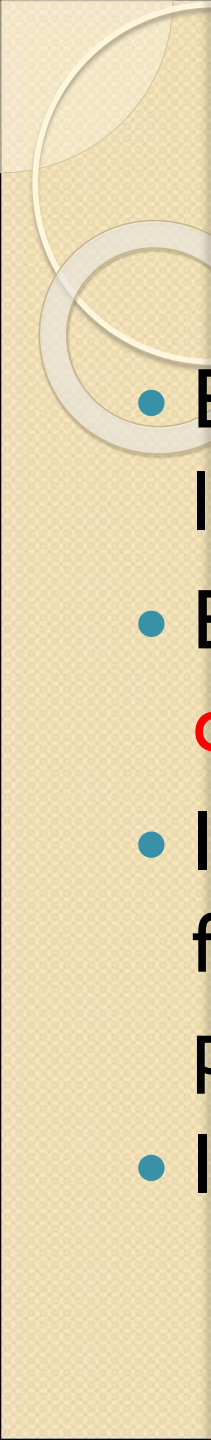






# Psychosocial Theory of Erik Erikson

- Erik Erikson proposed the psychosocial theory of development.
- It which suggests that an individual's personality develops throughout the lifespan based on a series of social relationships.
- This theory emphasized the social relationships that are important at each stage of personality development.

- 
- Erikson identified eight stages from infancy to late adulthood.
  - Each stage is characterized by a **psychosocial crisis** of two conflicting forces.
  - If an individual successfully reconciles these forces there is development of a healthy personality and a sense of competence
  - If not a negative outcome is generated

# Stages of Personality Development

## Stage I: Trust vs. mistrust ( 0–1 year) -Infancy

- Centers around the **infant's** basic needs being met by the parents and this interaction leading to trust or mistrust
- Infants develop trust with a warm response from the caretaker.
- the major developmental task in infancy is to learn whether or not other people, especially primary caregivers, regularly satisfy basic needs.
- If caregivers are consistent sources of food, comfort, and affection, an infant learns trust—that others are dependable and reliable.

- 
- If they are neglectful, or perhaps even abusive, the infant instead learns mistrust—that the world is an undependable, unpredictable, and possibly a dangerous place
  - Development of mistrust can lead to feelings of frustration, suspicion, withdrawal, and a lack of confidence.

## Stage 2 :Autonomy vs. shame and doubt ( 1–3 years)- Early Childhood

- Children at this age like to explore the world around them and they are constantly learning about their environment.
- The parents' patience and encouragement helps foster **autonomy** in the child.
- Children feel autonomous if caregivers encourage independence. Otherwise, they will doubt their abilities
- If caregivers demand too much too soon, refuse to let children perform tasks of which they are capable, or ridicule early attempts at self-sufficiency,
- children may instead develop **shame and doubt** about their ability to handle problems.

## **Stage 3: Initiative vs. guilt ( 3–6 years)- Middle Childhood)**

- By imaginative play, kids experiment with their ambitions.
- If parents do not encourage their initiative, the kids will feel guilt.
- The child is learning to master the world around him/her, learning basic skills and principles
- They learn how to zip and tie, count and speak with ease.
- If parents and preschool teachers encourage and support children's efforts, while also helping them make realistic and appropriate choices, children develop initiative—independence in planning and undertaking activities.



- But if, instead, adults discourage the pursuit of independent activities or dismiss them as silly and bothersome, children develop guilt about their needs and desires
- **Stage 4: industry vs. inferiority ( 6–12 years) –Late childhood)** Industry here refers to purposeful or meaningful activity. It's the development of competence and skills, and a confidence in carrying out a task.

- 
- In school, children learn to work as a group.
  - They will have inferiority feelings if the peer environment is hostile.
  - A child who experiences failure at school tasks and work, or worse still who is denied the opportunity to discover and develop their own capabilities and strengths and unique potential, is prone to feeling inferior and useless.

- Inferiority is feeling useless; unable to contribute, unable to cooperate or work in a team to create something, with the low self-esteem that accompanies such feelings.
- **Stage 5: Identity v Role Confusion ( 12-20)- Adolescent**
- Self-identity is a significant development during adolescence.
- Identity means essentially how person sees themselves in relation to their world.
- Role Confusion is the negative perspective ( an absence of identity) - meaning that the person cannot see clearly or at all who they are and how they can relate positively with their environment.

- Young people struggle to belong, accepted and affirmed, and yet also to become individuals.
- **Stage 6: Intimacy vs Isolation ( 20-40yrs)- Early adulthood**
- Those who cannot establish relationships or intimacy are prone to be socially isolated
- Intimacy means the process of achieving relationships with family and marital or mating partner(s).
- It is the giving and receiving of physical and emotional connection, support, love, comfort, trust,

- Isolation conversely means being and feeling excluded from the usual life experiences of dating and mating and mutually loving relationships. It is characterised by feelings of loneliness, alienation, social withdrawal or non-participation.
- **Stage 7: Generativity vs Stagnation( 40 -65)-Middle Adulthood**
- Generativity, then is primarily the concern in establishing and guiding the next generation... the concept is meant to include... productivity and creativity."

- 
- When a person makes a contribution during this period, perhaps by raising a family or working towards the betterment of society, a sense of generativity—a sense of productivity and accomplishment—results
  - E.g. Parenting is the best example to guide the younger generation.
  - In contrast, a person who is self-centered and unable or unwilling to help society move forward develops a feeling of stagnation—a dissatisfaction with the relative lack of productivity



## Stage 8: Ego integrity vs. despair (65 – death)- Late adulthood

- As we grow older and become senior citizens we tend to slow down our productivity and explore life as a retired person.
- We contemplate our accomplishments and are able to develop **integrity** if we see ourselves as leading a successful life.
- If we are not satisfied with what they did during their lifetime and see our life as unproductive, or feel that we did not accomplish our life goals, we become dissatisfied with life and develop **despair**, often leading to depression and hopelessness.

# James Marcia Theory of Identity Statuses

- He extended the **Identity vs. Role Confusion** stage of Erikson's theory.
- He states that identity in adolescents results from a series of choices.
- These choices, regard to which ideologies, occupations, etc a person has incorporated into their identity, are affected by **conflict /crisis and commitment**.
- Marcia theorized that the concept of identity is made up of four identity stages that include

# Identity diffusion

- Individuals in this part of the development of identity usually have **not had a crisis and have no commitment to anything.**
- They have made no effort to explore about or experience various options.
- They show little interest and are undecided about occupational and ideological choices.
- For example, they may be having too much fun after form 4 to decide whether to attend college or think of future goals.

# Identity foreclosure

- Individuals in this part of development usually have not had a crisis yet but have a high commitment to a life path.
- A decision is made without looking into alternatives.
- This happens when parents hand down commitments and choices to their adolescent usually on an authoritarian manner e.g. join their parents in the family business
- Therefore, have no opportunity to explore different approaches, vocations, ideologies on their own.

# Identity moratorium

- Individuals in this area of development have experienced a crisis or are going through one but have yet to commit to an identity.
- They experiment with multiple options in order to discover the best choice for them.
- i.e. they know they ought to have decided which course /career to take but are undecided between many options ( law, medicine, accounts).
- Such people are therefore in a stagnation/inertia state.

# Identity achievement

- It describes people who have undergone a crisis and have committed to an identity.
- Commitment is high since the person has explored many options
- e.g. during high school, Peter discussed extensively career opportunities with the guidance counsellors who helped him to become aware of his exceptional abilities in science and maths hence made a choice to study engineering.



Has a crisis been experienced?

Has a  
commitment  
been made?

|     | Yes                     | No                    |
|-----|-------------------------|-----------------------|
| Yes | Identity<br>achievement | Foreclosure           |
| No  | Moratorium              | Identity<br>diffusion |

# SOCIAL DEVELOPMENT

- **Social development** is the process through which we *acquire*
- *the ability to behave in accordance with social expectations,*
- *play approved social roles,*
- *develop appropriate social attitudes and*
- *acquire a sense of self-identity*

# Theories of Social Development

- ***The Psychoanalytic theory-***
- Social development is influenced by the development of sex typed behaviors.
- These are acquired when children learn to identify with the same sex parents after resolving the Oedipus and Electra complexes.

- 
- ***Social learning theory***- children develop socially by observing and imitating others in the immediate environment e.g. parents, siblings, teachers, peers.
  - Appropriate imitation is reinforced e.g. when boys display aggressive behavior and girl's submission, dependence, it is reinforced.
  - Inappropriate behaviors are punished reducing their chances of being repeated.

- 
- **Cognitive theory** – emphasizes the role played by mental processes in understanding one's sex and gender e.g. gender identity begins when one notices the anatomical differences between boys and girls.
  - The child then labels themselves male/female and begins to act accordingly.
  - If this doesn't happen one may experience gender identity problems such as transsexuals.

# The process of social development

- **Birth to 6 months** – neonate is insensitive to people, self-centered, little communication
- **6-24 months** – child begins to appreciate the caregivers (feeding, care, warmth)
- **2-5 years** – child is gradually separated from the mother and starts interacting with siblings, other children, school
- **6-11 years** – spends more time in school with peers of same sex
- **12 years onwards** – adolescence sets in and peer group becomes very important as they begin to associate with members of opposite sex.



# ***Developmental stages of play***

- **The developmental stages/styles of play include**
- ***Solitary play/ non-social play*** (2-3 years) – the child plays alone, ignoring other children nearby but may observe them as they play
- ***Parallel play*** – the child plays along side of other children but does not share or cooperate with them
- ***Associative play*** (4-5 years) children play with others in a common activity, share materials but there is no coordination, division of labour or common unified goal- each is doing their thing

- 
- **Cooperative play**- more organized with common goals and rules and so children discuss plans and assign roles
  - **Organized or group play** – elaborate competitive games with rules are played and children enter the world of organized sports

# Types of Play

- ***Functional play (1-3 years)*** – common with boys, repetitive muscular activities with or without objects e.g. pulling, pushing toys, climbing, harassing things
- ***Constructive play (4-6 years)*** – manipulation of objects to build/create things e.g. clay, blocks. Most common with girls and helps in problem-solving by increasing flexibility and consolidates learning, enhances creativity

- ***Dramatic play/pretend play (6-7 years)*** – use of imaginary situations and characters e.g. use of costumes to role play, rehearse problems in a no failure situation, use of fantasy
- ***Games for older children (7-12 years)*** – games with structures, rules and goals and involve a lot of cooperation in large or small groups. Teaches children how to win or loose

# Agents of socialization

## Definition of Agent of Socialization

- These are significant individuals, groups, or institutions that influence our sense of self and the behaviors, norms, and values that help us function in society. They include
- **Family**-train children in eating habits, toilet training, sex modesty, and dressing. Tidiness, respect for adults and authority figures, cultural inhibitions. All this is done through rewards, punishments and imitation of role models.
- **Schools** – how to relate to others formally and informally, knowledge and skills, proper attitudes to help fit in society, values of society reinforced e.g. cleanliness, respect, cooperation
- **Church**- virtues( Godly behavior), values and rules emphasized.

- **Peer groups** – children learn meaning of roles, status, leadership, influences fashion, habits, trends etc. **Importance of peers:**
- They provide opportunity to practice leadership skills which may in turn enhance self-esteem necessary for a healthy personality development.
- Teaches children variety of social skills e.g. sharing, cooperation
- Provide emotional outlet and social emotional support when one has problems
- Allows for free expression among equals without fear or judgment or reprimands



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- ***Social organizations*** e.g. clubs, societies
  - ***Mass media*** – electronic and print media which educates, informs, influences. It can enhance prosocial behavior, language and cognitive development, anti-social behavior e.g., aggression, gender, racial stereotypes.

# Importance of play to children

- Learns about himself and others
- Learns about the world and how he/she fits into that world
- Develops skills, solves problems and explores social relationships
- Learns to communicate with others, get new information and understanding, gain vocabulary
- Helps children to preserve and build attention skills, listen to others views, sympathy, endurance
- Stimulates intellectual growth and develops mental skills, creativity

- 
- Is provided with practice in gaining a sense of autonomy and competence
  - Learns how to manipulate and control objects
  - Play helps children develop a sense of self-direction, trust in oneself, and a feeling of self-worth
  - Play allows children to release energy and gain body control, build muscular coordination, refine motor skills, tolerance
  - It can help children develop a conscience – to value fairness, honesty and good sportsmanship
  - Play allows children to work out tension and stress and to express inner feelings

- 
- It helps children **in their social-emotional** development and socialization process: learning his own feelings and learning the importance of understanding others' feelings,
  - sharing, taking turns and cooperating,
  - allows them to get rid of aggression and control it, builds friendships and social skills, provides safe environment for children to compare their own behavior and skills with those of other children

# Emotional development

- Emotional development involves learning what feelings and emotions are,
- understanding how and why they occur, recognizing your own feelings and those of others,
- and developing effective ways for managing those feelings appropriately

# Stages of Emotional Development

- **I. Noticing emotions: Birth to one**
- There are a lot of different theories about how emotions develop and function. Some think we are born with only three emotions: happiness, anger and fear. Others believe that babies are capable of feeling a much wider range from birth. It's impossible to know for sure when they can't tell us, but through crying and cooing, babies certainly communicate something. At this stage, an infant is discovering the world—good things like cuddles, bad things like full diapers. They are noticing how everything makes them feel.



## **2. Expressing emotions: 2-3years**

- As children develop a vocabulary and more independence, they will experiment with expressing emotion in new ways.
- Some of it will be productive like drawing and narrating a picture of the scary monster under the bed.
- Some of it will be more like throwing a tantrum in the grocery store because they can't get what they want.
- This can be a very difficult stage for adults as children experience complex emotions but have not yet figured out healthy versus unhealthy expression.



- **3. Managing emotions: (3-5years)**

- At this stage, children are ready to enter preschool.
- A new social environment and more independence provides a great opportunity for growth but also poses some new challenges.
- Sharing, listening and playing together can cause friction between children, and since they cannot rely on their parents all the time anymore, they must develop new coping skills to manage on their own.
- Preschool caregivers play a vital role in this development as they create a safe space and offer guidance

# SIX BASIC EMOTIONS



Anger



Happiness



Surprise



Disgust



Sadness



Fear



Angry



Disgust



Sad



Joy



Surprise



Fear

- **Emotional attachment**
- ***Attachment refers to the emotional bonds that we develop with those with whom we feel closest, and particularly the bonds that an infant develops with the mother or primary caregiver***
- ***Refers to formation of an enduring solid-emotional relationship between an infant and another person***
- First attachment is usually between infant and mother/caregiver.

- **Qualities of attachment:**

- ***Secure attachment***

- infant explores while alone with the mother and is visibly upset by separation
- when mother returns, infant is happy and welcomes physical contact with her

- ***Insecure attachment (anxious and resistant)***

- Child anxious, unlikely to explore while mother is present
- Distressed when mother departs
- Ambivalent/confused when mother returns i.e. unlikely to initiate contact

- ***Insecure attachment (anxious and avoidant)***

- Not interested in exploring when alone with mother
- Little or no distress when separated from mother
- Avoids contact when she comes back
- Scared of strangers



- **Stages of attachment**

- **1) Asocial** stage (0-6 weeks) – few protest strangers
- **2) Indiscriminate attachment** (6 weeks – 7 months)
  - -Will protest when put down/left alone
  - -Enjoy human contact indiscriminately (attention from everyone)
- **3) Stage of specific attachment** (7 months)
  - protest if separated from a particular person usually mother/caregiver
  - begin to fear strangers meaning genuine attachments have been established
- **4) Stage of multiple attachments** e.g. too many significant others in their lives e.g. fathers, (bonding) baby sitter, neighbors



- 
- **Factors promoting attachment**
  - Warmth and contact
  - Baby's appearance (chubby cheeks, soft rounded features, cute, loveable babies)
  - Ability to discriminate familiar persons from unfamiliar
  - Inborn reflexes that promote interaction e.g. rooting, smiling, grasping
  - Feeding responsiveness hence trust
  - If parents had planned for the baby

- 
- **Factors that inhibit attachment**
  - ***Unlovable babies*** – unresponsive, withdrawn (those born to drug addicts)
  - ***Unreachable/ unresponsive care givers*** especially if pregnancy was unwanted hence resentment, insecurity
  - ***Environmental factors*** e.g. a very big family so some children are neglected; depression due to unhappy marriages, lack of response/warmth to children

## Effects of attachment on later development

### ***a) Securely attached infants:***

- They are more curious, interested in learning, cooperative, friendly hence liked by teachers and peers, solve problems better
- Become social leaders, often initiate activities, self-directed, eager to learn, and are less dependent on adults.
- Are more popular with peers
- They derive comfort from close companions, whom they use as a base for exploration

- 
- Children who are well secured with nurturing adults develop good relationships with others.
  - Securely attached infants feel confident and are actively involved with others.
  - Securely attached infants have more varied vocabulary and more positive interaction with peers and more joyful and friendly
  - Secure attachments prepare infants for intimate and stable friendships in the other stages of development i.e. childhood, adolescence and adulthood

- ***b) Insecurely attached infants:***
- Insecurely attached infants do not venture far, are usually aggressive, and rejected by peers.
- Insecurely attached infants in schools tend to be socially and emotionally withdrawn, and hesitate to engage in play.
- Less curious, less interested in learning,, and less forceful in pursuing their goals
- More likely to be overly dependent, and to experience problems in school.
- Manifest more negative emotions such as fear, distress and anger.
- Display hostility, often have inhibitions, and develop behavior problems at all levels of schooling and psychiatric disorders.

# Emotional Development among Adolescents

- **Adolescents are characterized by heightened emotions.**
- This can be attributed to the following factors:
- ***Volatile parent- adolescent relations:*** parents are sometimes excessively domineering, and impose unreasonable restrictions on the adolescents.
- Adolescents on the other hand view their parents as old fashioned and this generates friction. Adolescents get confused particularly when reprimanded for making childish decisions.
- ***Adjustment at home and school:*** this is on account of additional responsibilities, new patterns and habits, more independence-which create emotional tensions.
- ***Unfavorable relations at home:*** lack of proper guidance especially on the problems of adolescence, and feeling misunderstood with regard to their interests and opinions/views.



- ***Societal expectations:*** adolescents are expected to behave like adults while physically and intellectually, they are not yet ready.
- ***Adjustment to members of the opposite sex:*** adolescents are in the process of developing heterosexual relationships, but are unable to understand the correct social behaviour.
- ***Religious conflicts:*** adolescents are very critical of their parent's beliefs, and need to determine/identify their own beliefs.
- ***School failure:*** they dread school failure, and can result to suicide, withdrawal from school, and even leaving home.
- ***Vocational problems:*** they worry about their future occupations, and are anxious when they see the state of unemployment in the country.

- 
- **Importance of emotional development**
  - Understanding and interpreting people's emotions, expressing one's emotions enhances social development.
  - Helps people enjoy good relations with peers/better in social competence.
  - Social referencing enables children to acquire knowledge e.g. a mother's pained expression at the sight of a child holding a knife may suggest the knife is dangerous.

- 
- Promotes social contact and helps caregivers to adjust their behaviour to the urgent needs and goals e.g. cry is a call for attention, anger means stop, smile/laugh means continue what you are doing.
  - Helps the child to understand the world he/she lives in (information from caregiver's emotional display).

# **How can educators support emotional development in children and young people?**

- Provide effective support for children and young people's emotional development starts with paying attention to their feelings and noticing how they manage them.
- Help children and young people recognize and understand emotions
- Tune in to children and young people's feelings and emotions

- 
- Set limits on inappropriate expression of emotions
  - Be a role model: Showing children and young people different ways you understand and manage emotions helps them learn from your example. If you lose your temper, apologise and show how you might make amends.

# MORAL DEVELOPMENT

- **Moral Development:** It is the process through which an individual acquires the ability to distinguish between right and wrong
- **Theories of Moral Development**
- **Piaget's Theory; Has three broad stages**
- **Stage I: Premoral ( birth – 5 years):** Children show little understanding of social rules. They invent their own rules; change rules at will



## **Stage2: Heteronomous morality or moral realism**

- **The children believe the rules from authorities are fixed, sacred and absolute.**
- **Believe justice is subordinate to adult authority. Eg its wrong to tell a lie because the parent say so.**
- **They also judge actions by looking at the consequences rather than intentions eg breaking more cups**

- 
- **Stage 3: Autonomous Morality or Moral relativism.**
  - **They judge whether s/thing is right or wrong by looking at intentions or consequences. They see rules as not static ie thru consensus they can be changed.**

# ***Kohlberg Theory of Moral Development.***

- **Kohlberg identified three levels and six stages of moral development**
- **Level I: Pre-Conventional**
- **Reasoners at this level judge the morality of an action by its direct consequences.**
- **Stage I. Obedience and punishment orientation(*How can I avoid punishment?*)**
- **Individuals focus on the direct consequences of their actions on themselves**

- For example, an action is perceived as morally wrong because the perpetrator is punished. "The last time I did that I got spanked, so I will not do it again."
- The worse the punishment for the act is, the more "bad" the act is perceived to be. [1]
- Stage 2. Instrumental- relativistic orientation
- Self-interest orientation(*What's in it for me?*)(*Paying for a benefit* [6])

- This reasoning shows a limited interest in the needs of others, but only to a point where it might further the individual's own interests.
- As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "You scratch my back, and I'll scratch yours" mentality.

## Level II: Conventional Level

- To reason in a conventional way is to judge the morality of actions by comparing them to society's views and expectations.
- Stage three: Interpersonal Concordance of good boy/ girl orientation
- good intentions as determined by social consensus, the self enters society by conforming to social standards.



- the need for individual approval .Individuals are receptive to approval or disapproval from others as it reflects society's views.
- They try to be a "good boy" or "good girl" to live up to these expectations, <sup>[2]</sup> having learned that being regarded as good benefits the self.

- 
- **Stage 4: Authority and social Order maintaining Orientation**
  - *ideal or ideals often prescribe what is right and wrong.*
  - *If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules.*
  - *When someone does violate a law, it is morally wrong;*
- SNR. OF HANCA MAXWELL PAUL

# Level III: Post-Conventional

- It is the principled level, marked by a growing realization that individuals are separate entities from society,
- and that the individual's own perspective may take precedence over society's view; individuals may disobey rules inconsistent with their own principles.
- view rules as useful but changeable mechanisms—ideally rules can maintain the general social order and protect human rights.
- Rules are not absolute dictates that must be obeyed without question.

- Stage five : Social contract Legalistic orientation
- the world is viewed as holding different opinions, rights, and values. Such perspectives should be mutually respected as unique to each person or community.
- Laws are regarded as social contracts rather than rigid edicts. Those that do not promote the general welfare should be changed when necessary to meet “the greatest good for the greatest number of people”.
- This is achieved through majority decision and inevitable compromise..



- **Stage six : Universal ethical principles**  
**Orientation**

- moral reasoning is based on abstract reasoning using universal ethical principles.
- Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. <sup>1</sup>
- In this way action is never a means but always an end in itself; the individual acts because it is right, and not because it avoids punishment, is in their best interest, expected, legal, or previously agreed upon.

# Factors influencing moral development

- Cognitive development- Decentration etc
- The family-direct instruction, role model, enculturation
- Social interactions esp with peers
- The school- direct instruction, modelling
- Religious organizations- teach morals,
- Mass media- models, information
- Community/culture- morals, norms, regulations, customs, laws etc.



# ADOLESCENCE PHYSICAL GROWTH AND DEVELOPMENT

- The onset of adolescence is heralded by two significant changes in physical development:
  - ❖ The adolescent growth spurt, that is a rapid acceleration in weight and height that marks the beginning of adolescence.
  - ❖ Puberty - derived from the Latin word “pubertas” which means age of manhood or literally to grow hairy. It refers to that point in life when we reach sexual maturity and become capable of producing a child.
- The period marked by the onset of the maturation of the reproductive functions is referred to as **pubescence**. It lasts for two years ending in puberty.

# ADOLESCENCE

- Adolescence is the period of development that begins at puberty and ends at emerging adulthood.
- The typical age range is from 12 to 18 years,
- This stage of development has some predictable physical milestones.
- **Physical Changes of Puberty**
- Puberty is the period of several years in which rapid physical growth and psychological changes occur, culminating in sexual maturity.

# Primary Sex Development

- These are changes related to the maturation of the external and internal sex organs.
- **In males**, the penis, scrotum, testes, seminal vesicles and the prostate gland mature.
- The presence of sperms in the male's urine is considered the principle sign of sexual maturation.
- **In females**, the ovaries, fallopian tubes uterus, vagina, and breasts also reach maturity. The most dramatic sign of sexual maturity in girls is the “**menarche**” i.e. the onset of menstruation( eleven to sixteen years).
- It occurs about two years after breast and uterine growth begin to grow and after a girl's height has slowed down.

# Secondary Sex Characteristics

- These are the features that accentuate the anatomical distinction between boys and girls.
- In males the voice becomes hoarse, wet dreams, hair in armpits and genital areas, facial hair, broadening of shoulders etc.
- In females, growth of breasts, widening of the pelvis (hips) and pubic hair.

# Early and Late Maturation

- **BOYS**

- Early maturing boys are more poised, relaxed, good natured, popular with peers, likely to be leaders and less impulsive than late maturers. They also have a high self-esteem.
- Early maturers are rated as physically more attractive, more composed, and more socially sophisticated than late maturers. They are independent, self-controlled and dominant.
- Early maturing boys may have problems in living up to others' expectations. They are expected to act mature than they are capable of.

- 
- **Late maturers** are said to be more dependent, more aggressive, and insecure. They are more likely to rebel against parents and think less of themselves.
  - They may feel and act more childishly.
  - Late maturers have the advantage of having a longer childhood whereby, they don't have to deal with the new and difficult demands of adolescence.



- **Girls**
- **Early maturing** girls have been found to be less sociable, expressive, poised and more introverted and shy.
- Early maturing girls make better adjustments in adulthood.
- **Late maturing** girls are initially more gregarious, socially poised, assertive, active, more popular with peers, and more satisfied with their body image.
- However, at late adolescence, the early maturing girl becomes more popular with both sexes, more self-poised, better at coping, more self-directed cognitively, socially and emotionally.

## Cognitive Development at the Adolescent stage

- Most of the adolescents are **at the Formal Operational stage** although some may still be at the concrete operational stage. **Characteristics of thinking at this stage are:**
- They can reason about abstract ideas. They are freed from concrete reasoning. They can think about the future, and can think in terms of symbols and imaginary events.

- They acquire the ability to generalize facts - deductive reasoning, see relationships and to solve problems of increasing complexity and difficulty.
- They have the ability to reason hypothetically, e.g. what if; supposing... They can generate very unusual and creative responses.
- They have the ability to appreciate metaphoric meaning, i.e. using a word or phrase with one literal meaning to describe another object or event. For example, “a blanket of snow”. “Shattered faith”.

- 
- Their approach to problem solving becomes increasingly systematic and abstract, like the hypothetical deductive reasoning of a scientist.
  - They have a tremendous development of long-term memory and imagination.
  - They can identify with conditions and characters in the larger world.
  - Argumentativeness. They have the desire to practice their newfound mental ability - to consider various points of view. It helps to stretch out their reasoning ability. They need to be engaged in discussions.

# Adolescent Egocentrism

## Forms of adolescent Egocentrism

### **Personal Fable.**

- Adolescents see themselves as much more central and significant on the social stage than they actually are.
- For example, they believe that no one else has ever had the particular emotional experiences they themselves are having, e.g. so angry, so much in love etc.
- An example is a teen telling the mother “ you don’t know how it feels to be in love”.
- They also see themselves destined for great fame and fortune e.g. discovering a cure for cancer or being a great philosopher



- **Imaginary Audience - “all eyes on me”**

- They agonize over the fact that others are constantly watching and making comments about them.
- They fantasize how others will react to their appearance and behavior. For instance, they will spend hours before a mirror thinking others will judge the final result.
- They will enter a room regarding themselves as the most attractive and admired human being alive.
- If they have a slight blemish, they will not want to go to school, or they will wish they were invisible.





- **Invincibility Fable.**

- The feeling that they are somehow immune to the laws of mortality and probability.
- They believe they have a mission in life and problems cannot befall them yet.
- They feel magically protected from harm such as death, accidents, pregnancies etc.
- This leads to risk taking behaviors e.g. drunken driving, promiscuity etc.

# Naive Idealism

- Inability to differentiate the ideal from the real/practical.
- They imagine an ideal world and they feel compelled to try and bring reality closer to the ideal.
- For example, they have perfect answers to social problems.

# Adolescent Moral Development

- According to Kohlberg, most adolescents are at the conventional level (stage three and four), but a small percentage especially the delinquents are at the pre-conventional level..
- Most adolescents are at stage three, which is characterized by mutual and interpersonal expectations, relationships and conformity.
- What is right is living up to what is expected by people close to you, which generally involves conformity to stereotypical good behavior.
- Anticipation of approval or disapproval by significant others and the need to be a good boy/girl guides behavior.
- At stage four, what is right is fulfilling the actual duties to which you have agreed.
- Right is maintaining the society, group or institution. One does the right to avoid the breakdown of the system or the institution “if everyone did it”.

# Emotional Development during the Adolescent stage

- The physical changes at puberty make their emotions change.
- Adolescent emotions have been described as heightened

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# Causes of Heightened Emotions

- **Parent - adolescent relations**
- Excessive parental domination; unreasonable restrictions; sibling rivalry etc.
- Feels parents are unreasonable and old fashioned. Leads to friction.
- They feel like children in decision-making, and when they become emotionally volatile, parents reprimand them for being childish.
- This makes the adolescent confused.

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- **Adjustment at home and at school**
  - The changeover to new patterns of habits creates emotional tensions.
  - The process of changing from total dependence to independence roles with greater responsibility disturbs the adolescent.



- **Unfavorable relations in the home**
- Lack of proper guidance in preparation for the problems of adolescence; lack of understanding their interests and points of view.
- **Social Expectations**  
They are expected to think and act like an adult for whom he is not physically and intellectually ready.
- **Adjustment to members of the opposite sex**
  - There is attraction towards the members of the opposite sex but the adolescent is not able to understand the correct social behavior

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- **Religious conflicts**
  - He critically examines his parent's beliefs and begins to criticize them
  - **School failure**
  - Failure in school causes heightened emotionality so much so that a number of adolescents commit suicide leave home or withdraws from school.
  - **Vocational problems**
  - They develop anxiety when they see many unemployed. They worry about their future occupation.

# CHALLENGES AND PROBLEMS FACING ADOLESCENTS.

- **Common adolescent problems include**

- juvenile delinquency
- suicidal tendencies
- substance abuse
- chronic aimlessness
- eating disorders
- truancy

# Challenges

- In addition, the adolescent is faced by a number of challenges, which include
- resolving of identity issues
- sexuality related challenges
- school and career-related challenges
- choice of friends
- peer pressure
- conflict with adults.

# **Ways to help the adolescent to cope with the challenges and problems of this period:**

- Provide a favorable home environment where there is unconditional acceptance and love, and respect. Parents also need to model responsible behavior and learn to listen to what the adolescents have to say. In addition, provide basic necessities.
- Develop accountability and responsibility in the adolescents. This can be achieved through discussion of conflicting values, giving them experience in decision-making, giving them opportunities to do things for themselves and letting them experience the consequences of their behavior.

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- Provide continuous guidance and counseling as the need arises. Prepare them for the changes before they occur. Give them information so as to make informed choices and decisions.
  - Encourage them to discover their talents and support them. Identify skills they can train in especially during school holidays.
  - Encourage delay of gratification and self-control. Let them participate in community service and other sublimation activities.



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- Parents and teachers to tone down their domineering attitudes. Involve them in decision-making and discussion of various issues affecting them. use of a more democratic process or the participatory approach.
  - Train them in **life skills** such as decision-making, assertiveness and choice making.
  - Encourage high self-esteem and positive self-concept.
  - Provide emotional support and intervention for those who may face problems. For example, provide suicide awareness programs, psychotherapy, drug therapy and peer counseling to those who may be in a crisis.

# DEVELOPMENTAL TASKS

- Developmental tasks are inescapable expectations or obligations that a person has to attain by the end of a specific stage of development.
- eight developmental tasks that have to be satisfied during the adolescent period:
  - Attaining individuality
  - Making progress towards an organized personality pattern
  - Developing a philosophy of life  
Developing a concept of values and desirable behavior
  - Achieving a place in society
  - Understanding personal assets and liabilities
  - Maturing of plans for future living
  - Establishing deep personal relationships with individuals of both sexes.

# Adulthood

- **Early adulthood**
- **Middle adulthood**
- **Late adulthood**
  - **Developmental Tasks -**
  - **Physical Characteristics**
  - **Cognitive Characteristics**

# EARLY ADULTHOOD (20 – 40 YEARS)

- **Developmental Changes**
- **Physical Characteristics**
  - Strength, energy and endurance characterize early adulthood.
  - The peak of **muscular strength** occurs at about 25-30 years.
- **Cognitive Characteristics**
  - The main feature of adult thinking appears to be its practical nature, which allows for intellectual contradictions to exist side by side in a person's mind in the furtherance of a practical goal.
  - Adults focus their intellectual energies on solving real problems.
  - They accept contradiction, imperfection and compromise as part of adult life.

# MIDDLE ADULTHOOD (41-65 YEARS)

- **Developmental Changes**
- **Physical characteristics**
  - A number of physical changes take place during this period.
    - There is a marked and gradual decline in many aspects of physical development.
    - Vision and hearing begin to decline.
    - Incidence of cardiovascular disease increases during this time.
    - Blood pressure usually rises and the coronary arteries that supply blood to the heart begin to narrow during this time.  
**These changes are linked to the lifestyle of a person.**
  - Muscular strength declines steadily during this period.

# Menopause

- It is defined as the end of menstruation, which occurs as a result of ovarian degeneration and a decline of **estrogen** secretion.
- It occurs over a prolonged period of time from around 45 – 65 years.
- Major changes /characteristics include: -
  - Aging of ovaries and decline in output of ovaries i.e. the ova, estrogen and progesterone.
  - This leads to inadequate uterine stimulation, which consequently leads to the end of menstruation and discomfort when having sex.
  - Hot flashes, headaches, dizziness, heart palpitations and joint pains.



# Male Climacteric syndrome

- Although men do not experience menopause, they experience age related changes in the reproductive system.
- Climacteric syndrome refers to the loss of the ability to reproduce.
- These changes appear from the 60's to the 70's.  
**Characteristics include: -**
- Gradual decline in sperm production although viable sperms are produced by the oldest men.
- Androgen (testosterone) levels gradually decrease and erection occurs more slowly leading to decreased sexual performance.
- The seminal fluid becomes thinner and its volume decreases.

# Midlife Crisis:

- This is a period of transition at about 40 to 50 years during which individuals take stock and a look at their lives and ask was it worth the trouble?
- They may feel effective and competent and at the peak of their powers.
- To others, it may be a painful process. It's usually caused by a shift in control. They accept that they have no control over factors such as disease, death etc.
- Factors that may facilitate the crisis are divorce, children leaving home, menopause, infidelity, competition, impotence, job changes, death of a spouse, biological aging and severe illness.

## LATE ADULTHOOD AND AGING (65 AND ABOVE)

- **Developmental changes**
- **Physical characteristics**
- There are several major changes at this stage.
- The entire circulatory machinery is less efficient
- Sensory and perceptual functions decline. They have visual problems, hearing loss, sense of taste declines, and vestibular senses loose efficiency and touch and movement decline with age.
- The skin looses its elasticity i.e. its folded and wrinkled. Hair becomes thinner and turns gray.
- They shrink in size and stoop. Chemical composition of the bones changes causing an increased chance in bone fractures.

# Cognitive changes

- Lack of mental alertness. Loss of memory and forgetfulness, which is made worse by cardiovascular disease and senility.
- Decline in creativity.
- More effective though slow in their work due to accumulated experience and knowledge.
- Has a large amount of wisdom.
- Should be encouraged to be active both physically and mentally.

- **Death and dying: Epilogue**
- **Typical reactions to impending death include: -**
- Denial
- Anger – directed at death, against those not dying, the healthy and robust, dependants, what is left undone etc. they should not be taken seriously.
- Depression – allow them opportunity to express their sorrow
- Bargains for additional time
- Acceptance
- Hope



END

SNR. OHANGA MAXWELL PAUL





**ASK A DOUBT**